Supplementary Information for Natural Capital and the Measurement of Productivity*

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*All errors are our own.

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1 The Theory of Natural Capital and Total Factor Productivity in Growth Accounting

This appendix examines the conditions under which incorporating a previously unmeasured natural capital input into a macroeconomic production function improves residual-based estimates of total factor productivity (TFP). Our laboratory for measurement is a theoretical economy in which output is governed by TFP as well as by the inputs to the aggregate production function F. We assume that F is log-linear in inputs and takes at least four factors as arguments: labor L, produced capital K, and $J > 1$ natural capital inputs indexed $\{N_{j=1}, N_{j=2}, \dots, N_{j=J}\}$.[1] The logarithm of output at each instant t is given by

$$\begin{aligned}\ln(Y(t)) &= \ln\Big(TFP(t) \times F(K(t), L(t), \{N_j(t)\}_{j=1}^{J})\Big)\\ &= \ln(TFP(t)) + \gamma_K(t)\ln(K(t)) + \gamma_L(t)\ln(L(t)) + \sum_{j=1}^{J}\gamma_j(t)\ln(N_j(t))\end{aligned} \tag{1}$$

where $\gamma_K, \gamma_L, \{\gamma_j\}_{j=1}^{J} > 0$ denote output elasticities with respect to the corresponding factor inputs. If F is homogeneous of degree one in its inputs (i.e., production exhibits constant returns to scale, CRS), output can be expressed as

$$Y(t) = TFP(t) \times \Big(F_K K(t) + F_L L(t) + \sum_{j=1}^{J} F_{N_j} N_j(t)\Big) \tag{2}$$

where F_x denotes the partial derivative of F with respect to argument x. Under the assumption that the output elasticities γ are time-invariant, the growth rate of output in (1) at a given time t can be expressed as[2]

$$\frac{\dot{Y}(t)}{Y(t)} = \frac{T\dot{F}P(t)}{TFP(t)} + \frac{\dot{F}(t)}{F(t)} = \frac{T\dot{F}P(t)}{TFP(t)} + \gamma_K\frac{\dot{K}(t)}{K(t)} + \gamma_L\frac{\dot{L}(t)}{L(t)} + \sum_{j=1}^{J}\gamma_j\frac{\dot{N}_j(t)}{N_j(t)} \tag{3}$$

where dots denote derivatives of a given variable with respect to time. Let $g_X(t) \triangleq \dot{X}(t)/X(t)$ denote the growth rate of variable X at time t for $X \in \{A, K, L, \{N_j\}_{j=1}^{J}\}$. Rearranging (3) yields the accounting-based formula for calculating TFP growth: g_{TFP} is output growth less an output-elasticity weighted sum of growth in inputs:

$$g_{TFP} = g_Y - \gamma_K g_K - \gamma_L g_L - \sum_{j=1}^{J}\gamma_j g_{N_j} \tag{4}$$

We refer to (4) as the *growth accounting* equation going forward. The growth accounting equation states that if output elasticities in the production function are known to the econometrician and

[1] Without loss of generality, other omitted inputs that are not natural capital can be subsumed into the set of J inputs beyond produced capital and labor.

[2] Equation (3) still holds when output elasticities depend on the level of inputs and/or time, but they must be time-indexed.

she observes growth in both inputs used and output, she can recover the full path of TFP growth.

Several barriers complicate the inclusion of natural capital (NK) stocks in empirical growth accounting exercises. First, a national accountant may not know the full set of natural capital stocks $\{N_j\}_{j=1}^{J}$ that enter the production function. National statistical agencies have not traditionally collected data on natural capital. The System of National Accounts (SNA) standard international guidance includes a set of natural assets, but few countries tabulate changes in these assets over time ([5]). The System of the Environmental Economic Accounts Central Framework and Ecosystem Accounts provides an alternative set of natural assets they recommend countries track [17]. While some countries have begun collecting their own natural capital accounts, these data are unlikely to capture the full scope of natural assets that may enter the production function. Second, the accountant cannot (at least directly) observe output elasticities $\gamma_{\cdot}$. Elasticities must be estimated, which introduces uncertainty when these elasticities are used in the growth accounting equation. Third, the accountant may face measurement error when calculating growth in output and factor inputs as well as any other auxiliary measurement required to estimate output elasticities.

This section lays out a theoretical framework for examining how uncertainty over which natural capital stocks enter the production function affects estimates of output elasticities and TFP growth. We show conditions under which knowing more information about some, but not all, inputs sharpens estimates of TFP growth. These are policy-relevant questions; it is unlikely that national accountants will go from current practices to identifying the complete set of NK assets without first identifying an incomplete set. Further, budget and institutional constraints are unlikely to allow for the introduction of all NK assets at once. The theory here outlines conditions under which an intermediate step measuring some of the NK inputs yields contemporary improvements in TFP measurement.

1.1 Estimating Output Elasticities

Estimating TFP growth using the growth accounting method requires estimates of the output elasticities in (4). There are two general approaches that require distinct sets of assumptions. The first approach, the income approach, relaxes the assumption of log linearity and operates under two alternative assumptions: F is instead assumed to exhibit constant returns to scale (as in 2), and that the market structures for factor inputs as well as intermediate and final goods are known. The second approach, the regression approach, requires only that the aggregate production function in the economy is log-linear as in (1). This section touches briefly on the income approach before giving an overview of the regression approach as well as its properties when natural capital is omitted from the growth accounting equation.

1.2 Income-Based Approach

When the production function F exhibits CRS and factor markets are perfectly competitive, the shares of national income accruing to each factor input are equal to their output elasticities in equilibrium ([2]). For the case of capital, this implies

$$\gamma_K = \frac{F_K K}{Y} = \frac{RK}{Y} \tag{5}$$

The ratio of total rents paid to capital, RK, divided by output Y, which is usually measured as gross domestic product (GDP), is an unbiased estimator for the capital output elasticity in the economy. This procedure is especially useful given that its arguments can be measured in *nominal* terms (i.e., using money units) as opposed to real terms. Analogous procedures can be used to estimate elasticities for labor and natural capital factors where the factor income accruing to them is observable.

These estimators become biased if any factor income attributable to an omitted factor (e.g., any natural capital stock that enters production but is not paid its factor income) is misattributed to observed factors. Manning, Taylor, and Wilen [14] illustrate how this case is likely for rival public goods such as fish stocks. The estimator in (5) is also biased if the aggregate production function does not exhibit CRS or the econometrician does not know the structure governing the market for factor inputs. The challenges of estimating factor shares using this method are well known [2, 12, 15]. The resulting measurement error when using the income-based approach to calculate TFP growth depends on how national income is measured, how national is attributed across the factors of production, and how the error propagates through to TFP calculations.

1.3 Regression-Based Estimators

The regression based approach relies on the log-linear restriction in (1). When production is Cobb Douglas or well-approximated by its Taylor expansion such as in the translog case, the first two moments of the stochastic process governing input and output growth is a sufficient statistic for the set of output elasticities. The regression approach remains valid even when markets are not perfectly competitive as long as *quantities* (as opposed to value products) of inputs and output are known.[3]

To fix ideas, consider the running example from the main text, an economy that uses two natural capital inputs, N_1 and N_2, with output elasticities γ_1 and γ_2 (in addition to produced capital and labor). Suppose a national accountant observes time series data of change in output, g_Y, (i.e., change in real GDP) along with growth in produced capital g_K and labor g_L and measures all quantities she observes without error. Assume from here forward for all factors she measures it is done without measurement error. An econometrician who observes g_K and labor g_L has sufficient data to estimate γ_K and γ_L using a regression model based on the presupposed data generating process

$$g_Y = \hat{g}_A + \hat{\gamma}_K g_K + \hat{\gamma}_L g_L + \epsilon \tag{Model 1}$$

where $\hat{g}_A$ is an intercept term that becomes an estimator of average TFP growth. However, equation

[3] This condition is admittedly rarely met in practice as output is typically measured in terms of monetary value as opposed to physical quantities.

(3), implies that the true data generating process (DGP) is

$$g_Y = \gamma_K g_K + \gamma_L g_L + \gamma_1 g_{N_1} + \gamma_2 g_{N_2} + g_{TFP} \tag{DGP}$$

If one estimates (Model 1) by ordinary least squares (OLS), the fitted coefficients will be subject to traditional omitted variable bias (OVB) ([10]). For example, when $\hat{\gamma}_K$ is the OLS estimand from Model 1 it identifies

$$\hat{\gamma}_K = \gamma_K + \gamma_1 \kappa_{N_1,\tilde{K}} + \gamma_2 \kappa_{N_2,\tilde{K}} + \kappa_{TFP,\tilde{K}} \tag{Bias 1}$$

where $\kappa_{X,\tilde{K}}$ is the coefficient from a linear projection of the growth of X onto the residualized growth in K. By (Bias 1) unless growth in capital, K, and in labor, L, are uncorrelated with growth in the omitted natural capital factor inputs and TFP, estimates of $\hat{\gamma}_K$ and $\hat{\gamma}_L$ from Model 1 will be biased. The magnitude of the bias depends on the magnitude of the output elasticities for omitted factors and on how they covary with observed factors.

Consider an alternative case where one of the productive natural capital stocks, N_1, is observed by the econometrician. The analyst may expect that including this additional factor will reduce the bias in the estimates. They consider estimating

$$g_Y = \tilde{g}_A + \tilde{\gamma}_K g_K + \tilde{\gamma}_1 g_{N_1} + \tilde{\gamma}_L g_L + \nu \tag{Model 2}$$

where tildes differentiate Model 2 from Model 1. The bias in the estimate of the capital output elasticity using (Model 2) is

$$\tilde{\gamma}_K = \gamma_K + \gamma_2 \lambda_{N_2,\tilde{K}} + \lambda_{TFP,\tilde{K}} \tag{Bias 2}$$

where the λs are coefficients from residualized auxiliary regressions as above. It is ambiguous whether (Bias 1) or (Bias 2) is greater without further assumptions.

Consider first the omitted variable bias when g_{N_1} and g_K are the only terms in the growth accounting equation that are not iid. Predictions of theory are immediate; OVB is present in model 1 only, and including g_{N_1} always (weakly) reduces bias when estimating γ_K. Now consider relaxing this assumptions slightly, such that $\rho(g_{N_1}, g_{TFP}) > 0$ while g_{TFP} remains uncorrelated with all other growth terms (including g_K).[4] Both models 1 and 2 now produce biased estimates of γ_K because $\kappa_{N_1,\tilde{K}}$ and $\lambda_{N_1,\tilde{K}}$ are nonzero, making it ambiguous which estimator is less biased. For Model 1, the bias is determined by $\kappa_{N_1,\tilde{K}}$, which is the correlation between g_{N_1} and the portion of the growth in capital that is not explained by growth in labor:

$$\hat{\gamma}_K - \gamma_K = \gamma_1 \kappa_{N_1,\tilde{K}} \tag{Bias 1$'$}$$

When the correlation between growth in N_1 and the growth in capital not explained by growth in other facts is positive ($\kappa_{N_1,\tilde{K}} > 0$), some of the growth in output caused by growth in N_1 is

[4] The choice of direction for the inequality here is arbitrary and symmetric results would follow if it were reversed.

misattributed to produced capital, leading to an overestimate γ_K. In Model 2, the bias is given by:

$$\tilde{\gamma}_K - \gamma_K = \lambda_{A,\tilde{K}} \tag{Bias 2$'$}$$

and will lead to an RMSE that depends on the magnitude of the correlation between growth in N_1 and growth in TFP. The bias in (Bias 2′) has the opposite sign of the correlation between growth in N_1 and TFP because the regression coefficient $\lambda_{A,\tilde{K}}$ is the covariance in TFP and the portion of capital growth *not* explained by g_{N_1}. Since g_K and g_{TFP} are *unconditionally* uncorrelated, as g_{TFP} is positively correlated with g_{N_1} and a portion of g_K that is correlated with g_{N_1}, the bias results from the negatively correlated with the portion of g_k that's orthogonal to g_{N_1}, which leads to an overall correlation that is zero. The result is that g_{TFP} is negatively correlated with $\tilde{K}$, the residualized portion of g_K, so that γ_K is underestimated when $\rho(g_{N_1}, g_{TFP}) > 0$.

1.4 Estimating TFP Growth Using Residuals

We now turn to examining how biases in estimated output elasticities propagate through to the estimation of TFP. This section begins with a general case where take estimated output elasticities as given before turning to the specific case where they are estimated using the OLS approaches described in Section 1.3 above. From here forward, we operate under the assumption that TFP growth g_{TFP} is an independent and identically distributed random variable.

For a national accountant with perfect information on output growth, output elasticities, and factor growth inputs, her estimand of interest when calculating expected TFP growth would be

$$\mathbb{E}[g_{TFP}] = \mathbb{E}[g_Y] - \gamma_K \mathbb{E}[g_K] - \gamma_L \mathbb{E}[g_L] - \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}] \tag{6}$$

and simply plugging the sample analogs of population moments on the right hand side of (6) would be a feasible and unbiased estimator of $\mathbb{E}[g_{TFP}]$.[5] However, as discussed above and in the main text, it is often the case that measurement of natural capital is incomplete.

Turning again to our running example, suppose a natural accountant only has access to data on growth in output, produced capital, and labor inputs. One option for estimating TFP is to use a version of the estimator in Model 1 which omits both forms of natural capital but accounts for the two factors she measures. With estimated output elasticities $\{\hat{\gamma}_K, \hat{\gamma}_L\}$ in hand, her estimator for average TFP growth $\mathbb{E}[g_{TFP}]$ is

$$\hat{g}_A \triangleq \bar{g}_Y - \hat{\gamma}_K \bar{g}_K - \hat{\gamma}_L \bar{g}_L \tag{7}$$

where bars denote sample averages. Let

$$\varepsilon \triangleq \hat{g}_A - \mathbb{E}[g_{TFP}] \tag{8}$$

[5] We assume all relevant moment conditions hold such that all estimands are well-defined.

be the difference between (6) from (7). Under the assumption she measures growth in produced capital and labor without error, taking expectations of (8) conditional on estimated output elasticities gives the conditional bias of the estimator in (7)

$$\mathbb{E}[\varepsilon|\{\hat{\gamma}_K, \hat{\gamma}_L\}] = [\gamma_K - \hat{\gamma}_K]\mathbb{E}[g_K] + [\gamma_L - \hat{\gamma}_L]\mathbb{E}[g_L] + \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}] \tag{9}$$

gives the bias of the average TFP growth estimator in (7).[6] The shorthand ε will ease notation going forward. Define the mean squared error (MSE) for this estimator as $\text{MSE}(1) \triangleq \mathbb{E}[\varepsilon^2]$.

Now, suppose the national accountant is also able to measure g_{N_1} without error and separately estimates a new set of output elasticities $\{\tilde{\gamma}_K, \tilde{\gamma}_L, \tilde{\gamma}_{N_1}\}$. If we augment equation (7) with an additional term capturing the contribution of N_1 to growth, our alternative estimator based on Model 2 is

$$\tilde{g}_A \triangleq \bar{g}_Y - \tilde{\gamma}_K \bar{g}_K - \tilde{\gamma}_L \bar{g}_L - \tilde{\gamma}_1 \bar{g}_{N_1} \tag{10}$$

Letting $\delta \triangleq \tilde{g}_A - \mathbb{E}[g_{TFP}]$, the conditional bias for the estimator in Model 2 is

$$\mathbb{E}[\delta|\tilde{\gamma}_K, \tilde{\gamma}_L, \tilde{\gamma}_{N_1}] = \mathbb{E}[\gamma_K - \tilde{\gamma}_K]\mathbb{E}[g_K] + \mathbb{E}[\gamma_L - \tilde{\gamma}_L]\mathbb{E}[g_L] + \mathbb{E}[\gamma_1 - \tilde{\gamma}_1]\mathbb{E}[g_{N_1}] + \sum_{j=2}^{J} \gamma_j g_{N_j} \tag{11}$$

where δ is again a useful shorthand going forward. The mean squared error for this estimator is defined as $\text{MSE}(2) \triangleq \mathbb{E}[\delta^2]$.

The magnitudes of biases and mean squared errors for each estimator depend in principle on both the estimated output elasticities as well as expected values of the growth rates of inputs. At this stage, it is ambiguous which estimator performs better under either criteria. We turn now to examining under what additional assumptions we can sign the bias in estimated TFP, and under what conditions including the first natural capital stock on the margin—shifting from Model 1 to Model 2—improves both of these criteria.

Our first result is formed under assumption that the production function displays constant returns to scale (like in 2). In this case, the output elasticities must sum to one. This implies that estimates of output elasticities used for the estimator derived from Model 1 must satisfy the identity $\hat{\gamma}_L = 1 - \hat{\gamma}_K$, and for Model 2 must satisfy $\tilde{\gamma}_L = 1 - \tilde{\gamma}_K - \tilde{\gamma}_1$. We may then state the following:

Proposition 1. When production exhibits constant returns to scale, if $\min_j\{g_{N_j}\} > \max\{g_K, g_L\}$, then $\mathbb{E}[\varepsilon] > 0$, and TFP growth is overestimated. If $\max_j\{g_{N_j}\} < \min\{g_K, g_L\}$, then $\mathbb{E}[\varepsilon] < 0$ and TFP growth is underestimated. In either case, adding an additional natural capital stock N_j to the

[6] Note equation above implicitly assumed that the expectations of $\bar{g}_L$ and $\bar{g}_K$ are independent of estimated output elasticities which is unlikely to hold in practice.

growth accounting equation unambiguously reduces bias as long as the econometrician estimates g_{N_j} and γ_j without bias. When neither case holds, the sign of the bias is ambiguous, even if all other elasticities and growth rates for included factor inputs are measured without error.

Proof. See Section 5.1. □

Even if the econometrician correctly imposes the assumption of CRS production and measures all factors and output elasticities without error, she cannot generally sign the bias induced by leaving out natural capital. Only under very restrictive assumptions on the growth rates of natural capital can she be sure of the sign of the bias and that inclusion of additional natural capital stocks is helpful on the margin. Otherwise, including an additional natural capital stock (going from model 1 to model 2) is not guaranteed to reduce the bias or mean squared error in her estimate even under very optimistic assumptions on measurement:

Lemma 1. Suppose an econometrician has perfect measurement of all growth rates and output elasticities for factor inputs she observes. Suppose further that she knows with certainty that growth rates for all factor inputs are independent and identically distributed. The effect of adding of an omitted natural capital stock to the growth accounting equation (7) is still ambiguous in terms of bias and mean squared error of the growth accounting-based TFP estimator unless it is the only remaining omitted natural capital stock.

Proof. See Section 5.2. □

In the case of perfect measurement of output elasticities and growth in factor use, the biases for TFP estimates from Model 1 and 2 are

$$\mathbb{E}[\varepsilon|(\hat{\gamma}_K = \gamma_K, \hat{\gamma}_L = \gamma_L)] = \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}]$$
$$\mathbb{E}[\delta|(\tilde{\gamma}_K = \gamma_K, \tilde{\gamma}_L = \gamma_L, \tilde{\gamma}_1 = \gamma_1)] = \sum_{j=2}^{J} \gamma_j \mathbb{E}[g_{N_j}]$$

respectively. As hinted at in the main text, only when all growth terms share the same sign in expectation is it guaranteed that $|\mathbb{E}[\delta]| \leq |\mathbb{E}[\varepsilon]|$. Without the all-the-same-signs condition the effect of adding natural capital stocks to (7) yields ambiguous effect on bias and MSE.

An alternative formulation for the bias from these estimators is to express the omitted as deviations from average growth across all natural capital factors that are omitted from a given model. Let $\mathcal{J}$ be the set of natural capital factors omitted under Model 1. The average factor shares and growth rates across these factors are

$$\bar{\gamma}^{\mathcal{J}} \triangleq \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \gamma_j \quad , \quad \bar{g}^{\mathcal{J}} \triangleq \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \mathbb{E}[g_{N_j}]$$

such that for Model 1 the bias is

$$\mathbb{E}[\varepsilon|\mathcal{J}] = J\mathbb{E}\left[\gamma_j \mathbb{E}[g_{N_j}]|\mathcal{J}\right] = J\left(\text{Cov}\Big(\gamma_j, \mathbb{E}[g_j]\Big|j \in \mathcal{J}\Big) + \bar{\gamma}^{\mathcal{J}}\bar{g}^{\mathcal{J}}\right) \tag{12}$$

and for Model 2, where included factors are set $\mathcal{J}'$ with cardinality $J-1$ the bias is

$$\mathbb{E}[\delta|\mathcal{J}'] = (J-1)\mathbb{E}\left[\gamma_j \mathbb{E}[g_{N_j}]|\mathcal{J}'\right] = (J-1)\left(\text{Cov}\Big(\gamma_j, \mathbb{E}[g_j]\Big|j \in \mathcal{J}'\Big) + \bar{\gamma}^{\mathcal{J}'}\bar{g}^{\mathcal{J}'}\right) \tag{13}$$

Proposition 2. Suppose that the variances of γ_j and $\mathbb{E}[g_j]$ are sufficiently small so that for any subset of natural capital stocks $\mathcal{J}'$ with cardinality $J-1$ the inequality

$$\frac{J}{J-1} \geq \frac{\bar{g}^{\mathcal{J}'}\bar{\gamma}^{\mathcal{J}'}}{\bar{g}^{\mathcal{J}}\bar{\gamma}^{\mathcal{J}}}$$

holds. Then when the covariance between γ_j and $\mathbb{E}[g_{N_j}]$ across capital stocks is zero, adding a correctly-measured capital stock and its output elasticity to the growth accounting equation lowers the magnitude of bias.

Proof. See Section 5.3. □

The assumptions on sign monotonicity required to ensure including a natural capital stock on the margin reduces bias can thus be weakened slightly in exchange for the assumption that the growth rate of natural capital inputs is uncorrelated with their output elasticities. For this to hold, individual natural capital stocks must grow in a similar manner to the ensemble average over time, the growth rates must not be too strongly correlated with their factor shares, and output elasticities and factor growth rates must be estimated without error. In practice, we view that even these weaker assumptions are unlikely to hold in the data.

2 Regression Methodology and Analysis

This section describes how we implement the regression model in Section 1 of the main paper that estimates the relationship between measured TFP growth and changes in natural capital stocks and at the country level.

2.1 Theory

When a national accountant measures productivity growth using the Solow-residual based method described in equation 3 of the main text, conditional on her choice of inputs $\boldsymbol{I}$ the estimator for TFP growth each period is

$$g_A(t)|\boldsymbol{I} = g_Y(t) - \sum_{i \in \boldsymbol{I}} \hat{\gamma}_i \hat{g}_i(t) = g_{TFP}(t) \; + \sum_{x \in \boldsymbol{X}} \gamma_x g_x(t) \; - \sum_{i \in \boldsymbol{I}} \hat{\gamma}_i \hat{g}_i(t) \tag{14}$$

where $\hat{g}_i$ and $\hat{\gamma}_i$ are her estimates of the growth rates and output elasticities of included inputs. When the Solow-residual based method is correctly specified such that $\boldsymbol{I} = \boldsymbol{X}$, (14) simplifies to

$$g_A(t)|\boldsymbol{I} = g_{TFP}(t) \; + \sum_{x \in \boldsymbol{X}} \gamma_x g_x(t) - \hat{\gamma}_x \hat{g}_x(t) \tag{15}$$

In our case in Section 1, there are two reasons to expect the second term on the RHS to be nonzero. First, even if we were certain we knew the seven renewable capital stocks measured in the CWON were the only ones contributing to production, we are still working with *estimated* growth rates $\hat{g}_x$ as opposed to observations without error. Second, the same logic applied to the underlying construction process for the factors that are included when a given measured series for TFP is constructed; the estimates of output elasticities and factor growth rates $\hat{\gamma}_x$ and $\bar{g}.$ that enter the PWT's growth accounting equation may also be measured with error.

This leads to the decomposition of the series we present in Section 1 of the main text. The specific TFP growth series we use from the Penn World Tables (PWT) version 10.01 ([7])—$RTFP^{NA}$, uses a version of (14) where natural capital services are not included in the set of included inputs $\boldsymbol{I}$. Assuming that the specification in the PWT captures all other factor inputs outside of natural capital (i.e., where the set $\boldsymbol{X} \setminus \boldsymbol{I}$ is comprised solely of natural capital inputs), (14) takes the form

$$g_A(t)|\boldsymbol{I}, \hat{\boldsymbol{\gamma}}, \hat{\boldsymbol{g}} = \underbrace{g_{TFP}(t)}_{\text{True innovation}} \; + \underbrace{\sum_{x \in \boldsymbol{X} \setminus \boldsymbol{I}} \gamma_x g_x(t)}_{\text{Omitted inputs}} \; + \underbrace{\sum_{i \in \boldsymbol{I}} [\gamma_i g_i(t) \; - \; \hat{\gamma}_i \hat{g}_i(t)]}_{\text{Measurement error in included inputs}} \tag{16}$$

Thus measured TFP growth at the country level (from the Penn World Tables or other sources) can be correlated with growth in a given natural capital stock in that country—g_{N_i}—through three channels: (1), correlation between the growth of true TFP (innovation, technical change) and changes in the use of N as an input; (2), resource N_i is an omitted productive input that is absorbed when g_A is constructed as a residual and (3), because of correlations between natural capital use and any measurement error in resources that are included in the original set of inputs $\boldsymbol{I}$ when g_A is constructed.

We wish to test the null hypothesis for channel (2)—that N_i is not an omitted variable—in isolation. As discussed in the main text, because we only observe g_A jointly as the sum of the three terms in (16), we are restricted to testing whether the correlations between growth in natural capital stocks and *measured* TFP are nonzero. In practice, we test (and reject) a joint null hypothesis that a proposed set of seven omitted natural capital stocks is uncorrelated with measured TFP growth g_A. Thus even when our results are statistically robust, it is possible that they are driven by underlying relationships between natural capital and TFP growth or measurement error as opposed to being omitted variables in the growth accounting equation. However, we believe such a result suggests that it is desirable to collect and organize the data systematically to resolve this question.

2.2 Data for Covered Countries in the Empirical Portion of the Main Text

We estimate equation (4) from the main text on a balanced panel of 117 countries over the 1996-2019 period. Table 1 displays the list of countries covered in the regression.

Table 1: Covered Countries

Angola	Argentina	Armenia
Australia	Austria	Bahrain
Barbados	Belgium	Benin
Bolivia	Botswana	Brazil
Bulgaria	Burkina Faso	Burundi
Cameroon	Canada	Central African Republic
Chile	China	Colombia
Costa Rica	Cote d'Ivoire	Croatia
Cyprus	Czech Republic	Denmark
Dominican Republic	Ecuador	Egypt
Estonia	Eswatini	Fiji
Finland	France	Gabon
Germany	Greece	Guatemala
Honduras	Hong Kong	Hungary
Iceland	India	Indonesia
Iran	Iraq	Ireland
Israel	Italy	Jamaica
Japan	Jordan	Kazakhstan
Kenya	South Korea	Kuwait
Kyrgyzstan	Laos	Latvia
Lesotho	Lithuania	Luxembourg
Macao	Malaysia	Malta
Mauritania	Mauritius	Mexico
Moldova	Mongolia	Morocco
Mozambique	Namibia	Netherlands
New Zealand	Nicaragua	Niger
Nigeria	Norway	Panama
Paraguay	Peru	Philippines
Poland	Portugal	Qatar
Romania	Russian Federation	Rwanda
Saudi Arabia	Senegal	Serbia
Sierra Leone	Singapore	Slovakia
Slovenia	South Africa	Spain
Sri Lanka	Sudan	Sweden
Switzerland	Tajikistan	Tanzania
Thailand	Togo	Trinidad and Tobago
Tunisia	Türkiye	Ukraine
United Kingdom	United States	Uruguay
Venezuela	Zambia	Zimbabwe

Table 2 shows the seven renewable capital stocks from the CWON that form our set of environmental inputs $\boldsymbol{N}$ along with the native quantity units of measurement and the coverage of the underlying data. Due to the paucity of data spanning the entire panel for some resources, we impute zeros to the growth rates for all resources where the observation in the prior year is a missing value. This imposes the assumption that there is no change in the true usage rate of natural services where data from the year prior is missing. The last column shows the share of data points that are imputed values of zero for each resource. Outside of mangroves, biomass, and hydropower, the imputation is minimal; however, for the latter three resources a huge portion of the original data are missing entries. To ensure our results are not driven by the imputation process, Table 4 shows that our results hold when we remove resources where a large fraction of observations are imputed and re-estimate equation 4 in the main text.

Table 2: Renewable resources used to estimate equation 4 in the main text

Resource	**Unit**	**% Obs. Missing**	**# Countries**
Urban land area	Index	0.00%	117
Total timber forest	Square Kilometers	1.25%	114
Agricultural land area	km^2	2.56%	116
Total forest-cover	Square Kilometers	3.95%	113
Mangrove ecosystem area	Hectares	63.25%	43
Total biomass	Metric tons	23.08%	90
Hydropower electricity generation	Gigawatt-hours	27.07%	86

2.3 Robustness Checks

We turn here to presenting a number of robustness checks of our empirical exercise in the main text. While we cannot test whether our results are driven by correlation between natural capital stock growth and objects (1) and (3) in equation (16), this section ensures that the results are not driven by our choice of outcome variables or included natural capital stocks.

2.3.1 Randomization Inference Tests of Joint Significance

While our estimating sample is a balanced panel, it's still possible that individual operations with high leverage are creating spurious results of joint significance. To check for this we perform a randomization inference test on the specification in column (3) of Table 1 in the main text. To operationalize this test, we resample growth in the seven natural capital stocks, $g_{N_{n,j,t}}$, within each country without replacement and randomly reassign them to years within the panel. This process, by construction, imposes that the correlation between these newly reassigned regressors $\tilde{g}_{N_{n,j,t}}$ and the outcome variable $g_{A_{j,t}}$ is expected to be zero within each country. We repeat this process 100,000 times. For each permuted dataset, we estimate the following regression

$$g_{A_{j,t}} = \sum_{n \in \boldsymbol{N}} \gamma_n \tilde{g}_{N_{n,j,t}} + \sum_{z \in \boldsymbol{Z}} \gamma_z g_{z,j,t} + \delta_j + \varepsilon_{j,t} \tag{17}$$

and perform a Wald test of joint significance for coefficients $\boldsymbol{\gamma}$ each time. Because the data are constructed such that the null hypothesis for the joint Wald test is true, this test forms an empirical distribution for $F-$test statistics under the null hypothesis. Figure 1 shows the empirical distribution of test statistics when the null statistic is true for 100,000 resampled versions of our panel. The p value from this test–the share of placebo F test statistics that are larger than the one in our estimating sample–is 0.015.

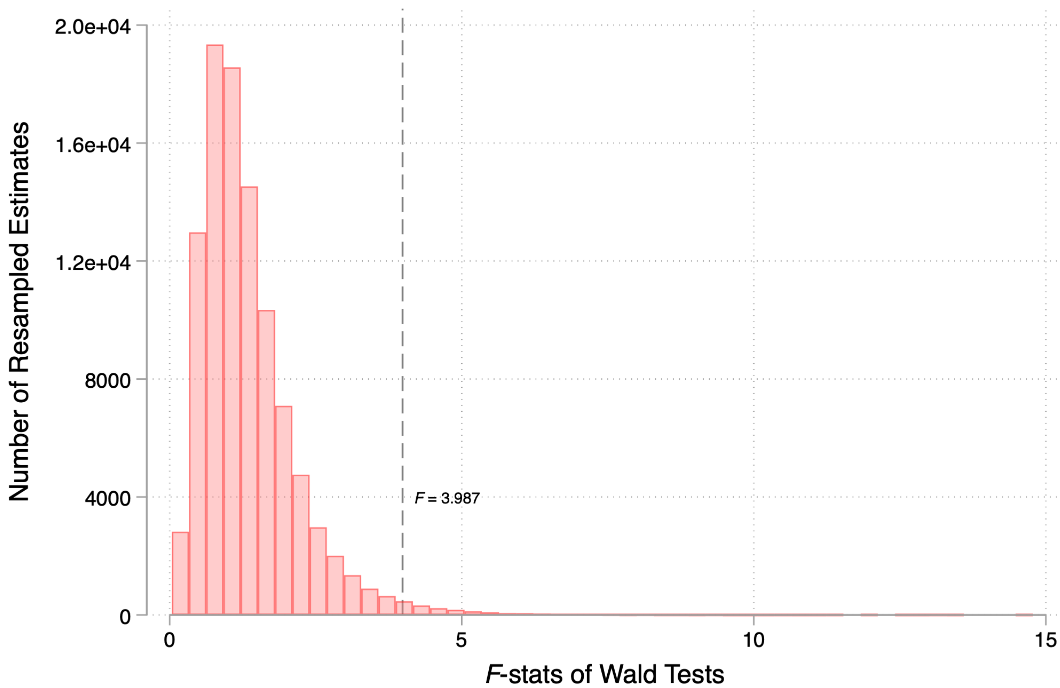

Figure 1: Histogram of F-statistics from Wald tests of joint nullity performed on 100,000 permuted samples of the data where the null is imposed true in expectation.

2.3.2 Covariate Choice

The seven natural capital stocks in Table 2 comprise the baseline set we test because these are the seven in the CWON dataset for which quantity data are public.[7] However, as the choice to include all seven is still somewhat arbitrary given they are ultimately only a subset of renewable natural capital, here we ensure that our rejection of the joint null result is not driven by our choice of regressors.

Excluding Urban Land Table 3 displays the results when equation 4 in the main text is re-estimated excluding urban land from the set of renewable natural capital stocks. The vast majority

[7]Data used to construct fisheries wealth in the CWON dataset are not made public.

of estimated coefficients remain positive for natural capital stocks exluding forest cover, in line with our results in the main text. Wald tests of joint nullity using asymptotic standard errors and semi-parametric methods lose some precision, but p-values still hover around traditional thresholds for statistical significance.

Table 3: TFP Growth vs. Natural Capital Growth Excluding Urban Land

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Timber Land	0.164	0.166	0.071	0.075	-0.003	0.026
	(0.090)	(0.075)	(0.074)	(0.071)	(0.068)	(0.104)
Agricultural Land	0.027	0.043	0.030	0.036	0.041	0.043
	(0.025)	(0.025)	(0.022)	(0.023)	(0.021)	(0.017)
Forest Cover	-0.067	-0.111	-0.005	0.012	0.083	-0.027
	(0.094)	(0.091)	(0.079)	(0.083)	(0.053)	(0.076)
Mangroves	0.122	0.067	0.357	0.327	0.318	-0.108
	(0.244)	(0.242)	(0.299)	(0.322)	(0.367)	(0.427)
Biomass	0.112	0.026	0.094	0.072	0.047	-0.104
	(0.098)	(0.077)	(0.108)	(0.101)	(0.135)	(0.108)
Hydropower	0.011	0.011	0.009	0.009	0.006	0.006
	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)
Controls for…						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
Wald	0.063	0.036	0.109	0.074	0.059	0.045
Bootstrap Wald	0.090	0.071	0.138	0.070	–	0.083
# Observations	2,808	2,808	2,808	2,808	2,808	2,457
# Countries	117	117	117	117	117	117

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables ([7]). All independent variables in the table enter the regression after being transformed into log changes. Rows under the "Controls for … " heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator. Standard errors clustered at the country level are shown in parentheses. Row "Wald" displays p values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Exclusion of Resources with Large Shares of Missing Observations Table 4 displays the results when equation 4 in the main text is re-estimated on a subset of four natural capital stocks excluding those where a large share of observations are either imputed or interpolated (see Table 2). We remain able to reject the joint nullity of estimated coefficients using asymptotic standard errors and semi-parametric methods at traditional levels.

Table 4: TFP Growth vs. Natural Capital Growth, Excluding Stocks with Large Portions of Missing Data

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Urban Land	0.295	0.298	0.198	0.209	0.146	0.062
	(0.092)	(0.096)	(0.065)	(0.073)	(0.061)	(0.115)
Timber Land	0.171	0.171	0.078	0.082	0.001	0.017
	(0.093)	(0.076)	(0.075)	(0.072)	(0.068)	(0.107)
Agricultural Land	0.029	0.045	0.029	0.035	0.041	0.046
	(0.025)	(0.025)	(0.022)	(0.022)	(0.021)	(0.017)
Forest Cover	-0.069	-0.114	-0.003	0.013	0.080	-0.028
	(0.097)	(0.092)	(0.079)	(0.083)	(0.053)	(0.076)
Controls for...						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
Wald	0.004	0.002	0.004	0.003	0.002	0.101
Bootstrap Wald	0.077	0.033	0.049	0.008	–	0.027
# Observations	2,808	2,808	2,808	2,808	2,808	2,457
# Countries	117	117	117	117	117	117

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables ([7]). All independent variables in the table enter the regression after being transformed into log changes. Rows under the "Controls for . . . " heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator. Standard errors clustered at the country level are shown in parentheses. Row "Wald" displays p values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Excluding Urban, Agriculture, and Timber Land Table 5 displays the results when equation 4 in the main text is re-estimated excluding all forms of land resources outside of forests from the set of renewable natural capital stocks. This will avoid any bias induced by land resources if they are correlated with measurement error in capital or labor income in national accounts that is used to estimate TFP growth in the PWT. Wald tests of joint nullity using asymptotic standard errors and semi-parametric methods become marginal or insignificant at around traditional thresholds for statistical significance under some specifications we consider. As discussed when estimating the specification in Table 4, some of this loss in precision may be due to the large share of missing values for factor growth we impute as zeros here.

Table 5: TFP Growth vs. Natural Capital Growth, Excluding Land

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Forest Cover	0.068	0.027	0.047	0.067	0.086	-0.013
	(0.061)	(0.062)	(0.068)	(0.075)	(0.043)	(0.070)
Mangroves	0.138	0.085	0.348	0.317	0.303	-0.096
	(0.240)	(0.237)	(0.298)	(0.322)	(0.366)	(0.428)
Biomass	0.114	0.031	0.092	0.070	0.039	-0.109
	(0.098)	(0.078)	(0.107)	(0.101)	(0.134)	(0.107)
Hydropower	0.011	0.011	0.009	0.009	0.006	0.006
	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)
Controls for...						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
Wald	0.050	0.061	0.111	0.107	0.062	0.229
Bootstrap Wald	0.062	0.101	0.117	0.120	–	0.191
# Observations	2,808	2,808	2,808	2,808	2,808	2,457

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables ([7]). All independent variables in the table enter the regression after being transformed into log changes. Rows under the "Controls for ... " heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator. Standard errors clustered at the country level are shown in parentheses. Row "Wald" displays p values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Inclusion of Non-Renewables The analysis in the main text looked at the correlation between TFP growth and changes in stocks of renewable resources as proxies for changes in services from natural capital. It is possible that the significant results there were due to omitted variable bias from leaving out non-renewable resources. Table 6 repeats the exercise in table 1 of the main text when we include all natural capital flows that can be calculated from the public version of the CWON dataset (i.e., renewable, fossil, and subsurface minerals). The full set of non-renewable and renewable natural capital stocks included are listed in Tables 11 and 12. Table 6 shows the estimated output elasticities associated with each natural capital stock when equation 4 in the main text is estimated on an unbalanced panel of 118 countries between 1972 and 2019 after expanding $\boldsymbol{N}$ to the full set of 20 natural capital stocks. Table 7 re-estimates this specification for the 1996-2019 period so it coincides with the time coverage in the main estimating sample.[8]

Overall, including the full suite of natural capital covariates available in the CWON dataset does not change the qualitative aspect of our results. Reassuringly, coefficients on oil and natural gas (resources where the associated industries comprise a substantial share of global GDP) are positive and significant at traditional levels in both tables. Urban land and hydropower remain precisely estimated. More generally, the coefficients on most natural capital stocks remain positive, in line with our priors that the use of non-renewable resources in production should be "goods" in terms of increasing output. While the coefficient on forest cover is negative, it is not significant at traditional levels and qualitatively aligned with some specifications shown in the main text. For these specifications including a broader suit of natural capital, we the null hypothesis to include all 20 natural capital stocks. Wald tests using both asymptotic and semi-parametric methods become substantially tighter around zero in all specifications we consider.

[8]The resulting number of observations and covered countries in this appendix does not coincide with that in the main text as here we adopt a more aggressive imputation strategy in order to fill in missing data. The process for constructing these quantity changes prior to 1996 is described in Section 3. Results are qualitatively unchanged when the estimating sample here is restricted to the 1996-2019 period, as shown in Table 7.

Table 6: TFP Growth vs. Renewable and Non-Renewable Natural Capital Growth

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Bauxite	0.002	0.002	0.001	0.001	0.000	-0.001
	(0.003)	(0.004)	(0.003)	(0.004)	(0.003)	(0.003)
Coal	0.005	0.007	0.009	0.009	0.007	0.004
	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)
Oil	0.014	0.014	0.016	0.015	0.015	0.015
	(0.007)	(0.007)	(0.007)	(0.007)	(0.008)	(0.008)
Copper	0.000	0.001	0.002	0.002	0.002	-0.002
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Phosphate	0.014	0.014	0.015	0.015	0.015	0.017
	(0.008)	(0.008)	(0.009)	(0.008)	(0.009)	(0.011)
Natural Gas	0.002	0.003	0.003	0.003	0.003	0.003
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)
Gold	0.002	0.002	0.003	0.003	0.002	0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Silver	0.001	0.001	0.002	0.002	0.002	0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)
Iron	0.002	0.003	0.001	0.002	0.003	0.004
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)
Tin	0.002	0.002	0.004	0.004	0.004	-0.000
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.005)
Lead	0.001	0.001	0.002	0.002	0.002	0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Zinc	-0.000	-0.000	-0.001	-0.001	-0.001	-0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)
Nickel	0.003	0.003	0.002	0.002	0.002	0.001
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Urban Land	0.294	0.307	0.225	0.239	0.249	0.099
	(0.080)	(0.093)	(0.061)	(0.068)	(0.074)	(0.078)
Agricultural Land	0.201	0.192	0.110	0.109	0.076	0.143
	(0.102)	(0.076)	(0.085)	(0.084)	(0.067)	(0.098)
Timber Land	0.015	0.034	0.037	0.046	0.042	0.037
	(0.025)	(0.025)	(0.023)	(0.023)	(0.021)	(0.018)
Forest Cover	-0.107	-0.171	-0.178	-0.206	-0.081	-0.084
	(0.105)	(0.091)	(0.095)	(0.095)	(0.090)	(0.104)
Mangroves	0.094	0.099	0.149	0.172	0.333	0.157
	(0.241)	(0.218)	(0.259)	(0.257)	(0.211)	(0.302)
Biomass	-0.101	-0.120	-0.020	-0.026	0.171	0.021
	(0.069)	(0.058)	(0.074)	(0.074)	(0.106)	(0.103)
Hydropower	0.013	0.013	0.012	0.012	0.010	0.010
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.004)
Wald	0.002	0.000	0.001	0.000	0.000	0.014
Bootstrap Wald	0.096	0.044	0.032	0.014	–	0.111
# Observations	5,255	5,255	5,255	5,255	5,255	4,901
# Countries	118	118	118	118	118	118

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables ([7]). All independent variables in the table enter the regression after being transformed into log changes. Rows under the "Controls for . . . " heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator. Standard errors clustered at the country level are shown in parentheses. Row "Wald" displays p values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Table 7: TFP Growth vs. All NK Growth, 1996-2019 Sample

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Bauxite	0.003	0.003	0.001	0.001	-0.001	-0.001
	(0.006)	(0.006)	(0.005)	(0.005)	(0.004)	(0.003)
Coal	0.004	0.006	0.002	0.003	0.003	0.001
	(0.006)	(0.005)	(0.004)	(0.004)	(0.005)	(0.004)
Oil	0.003	0.003	0.003	0.002	0.002	0.003
	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)
Copper	-0.003	-0.003	-0.003	-0.003	-0.004	-0.007
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)
Phosphate	0.018	0.017	0.021	0.020	0.020	0.020
	(0.011)	(0.011)	(0.013)	(0.013)	(0.014)	(0.013)
Natural Gas	0.000	0.000	-0.000	-0.000	-0.000	0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Gold	0.003	0.002	0.002	0.002	0.001	0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Silver	0.002	0.002	0.003	0.003	0.003	0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Iron	0.004	0.004	0.004	0.004	0.005	0.007
	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)
Tin	-0.002	-0.001	-0.001	-0.001	-0.001	0.001
	(0.002)	(0.003)	(0.002)	(0.003)	(0.003)	(0.003)
Lead	0.002	0.002	0.002	0.002	0.002	0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Zinc	-0.002	-0.002	-0.003	-0.003	-0.003	-0.004
	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.003)
Nickel	0.003	0.002	0.000	0.000	-0.000	-0.001
	(0.005)	(0.005)	(0.004)	(0.004)	(0.004)	(0.003)
Urban Land	0.311	0.311	0.209	0.219	0.152	0.170
	(0.092)	(0.097)	(0.065)	(0.073)	(0.060)	(0.064)
Agricultural Land	0.157	0.159	0.069	0.074	-0.009	0.147
	(0.090)	(0.076)	(0.073)	(0.071)	(0.065)	(0.089)
Timber Land	0.028	0.044	0.032	0.038	0.044	0.036
	(0.024)	(0.025)	(0.021)	(0.021)	(0.021)	(0.017)
Forest Cover	-0.067	-0.109	-0.014	0.004	0.083	0.075
	(0.095)	(0.091)	(0.081)	(0.083)	(0.053)	(0.103)
Mangroves	0.147	0.091	0.357	0.327	0.315	0.166
	(0.243)	(0.239)	(0.281)	(0.304)	(0.337)	(0.286)
Biomass	0.120	0.035	0.122	0.099	0.068	0.052
	(0.094)	(0.074)	(0.111)	(0.104)	(0.129)	(0.107)
Hydropower	0.011	0.012	0.010	0.010	0.008	0.010
	(0.005)	(0.005)	(0.004)	(0.004)	(0.004)	(0.004)
Wald	0.016	0.014	0.023	0.028	0.021	0.001
Bootstrap Wald	0.288	0.244	0.244	0.199	–	0.136
# Observations	2832	2832	2832	2832	2832	2802
# Countries	118	118	118	118	118	118

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables ([7]). All independent variables in the table enter the regression after being transformed into log changes. Rows under the "Controls for . . . " heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator. Standard errors clustered at the country level are shown in parentheses. Row "Wald" displays p values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null hypothesis. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

2.3.3 Alternative Productivity Measures

This section examines whether we find similar correlations between natural capital and productivity growth when we consider alternative measurements of TFP beyond those in the Penn World Tables. We consider two panels here: one from Eurostat covering the 27 countries in the European (plus Norway) which measures economy-wide multi-factor productivity ([4]), and a second covering only the agricultural sector produced by the Economic Research Service at the USDA ([8]).

Eurostat MFP Table 8 displays the results when equation 4 in the main text is re-estimated on multifactor productivity series generated by the European Statistical Agency (Eurostat) for the 27 EU countries along with Norway ([4]). The panel covers the years 2003-2019. Results from this alternative specification are shown in Table 8. While Wald tests of joint nullity using asymptotic standard errors remain significant, the semiparametric tests appear to lose some power.

Table 8: EU27 + Norway Eurostat MFP Growth vs. Natural Capital Growth

Outcome: $\Delta \ln(MFP_{j,t})$	(1)	(2)	(3)	(4)	(5)
Urban Land	0.524	-0.010	0.266	0.027	-0.134
	(0.498)	(0.369)	(0.495)	(0.397)	(0.481)
Timber Land	-0.477	-0.711	-0.818	-0.815	-1.147
	(0.363)	(0.328)	(0.262)	(0.274)	(0.475)
Agricultural Land	0.027	0.030	0.035	0.041	0.035
	(0.040)	(0.034)	(0.037)	(0.035)	(0.033)
Forest Cover	0.308	0.328	0.421	0.408	0.418
	(0.234)	(0.257)	(0.175)	(0.189)	(0.260)
Biomass	-0.443	-0.522	0.268	0.198	0.243
	(0.148)	(0.172)	(0.190)	(0.199)	(0.521)
Hydropower	-0.002	0.001	0.001	0.001	0.002
	(0.003)	(0.002)	(0.003)	(0.002)	(0.002)
Controls for...					
Factor Inputs	No	Yes	No	Yes	Yes
TWFE	No	No	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes
Wald	0.006	0.001	0.007	0.001	0.010
Bootstrap Wald	0.294	0.032	0.161	0.064	–
# Observations	476	476	476	476	476
# Countries	28	28	28	28	28s

Note: All independent variables in the table are measured in terms of log changes. The mangroves account is zero for all EU countries and is removed as a covariate for regressions in this table. Controls rows denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. Standard errors clustered at the country level are shown in parentheses. Row "Wald" displays p-values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

The results for the sign and magnitude of coefficients when estimated on EU MFP data differ substantially from the estimates using the full panel of TFP from the Penn World Tables in the main text. The coefficients on timber and forest cover swap signs; coefficients on timber land also become statistically significant at traditional levels. In contrast with this, urban land loses its statistical significance. To examine whether this difference in estimated coefficients is driven by heterogeneity across geographies within the main sample, we re-estimate our main specification on a subsample restricted to subset of EU countries plus Norway. This "apples-to-apples" comparison re-runs the regression in Table 8 using the Penn World Tables TFP series as opposed to the Eurostat MFP series as an outcome variable. To ensure a fair comparison, we restrict the sample to the time truncated time period of 2003-2019. During this period, the Pearson coefficient between the two measures of TFP growth is $\rho = 0.81$

Results for comparison are shown in Table 9. Inspection of the results when the outcome for the EU27 plus Norway is re-estimated using PWT data yields qualitatively similar coefficient changes. The signs on forests and timber land also flip. However, while the tables pass an eye test, the joint estimates of the coefficients are statistically different. When we estimate the specification in column (4) jointly for each outcome variable using a seemingly unrelated regression, a Wald test of the equality of all coefficients on natural capital between the two models rejects the null hypothesis of equality far beyond standard levels ($p < 0.001$).

Table 9: EU27 + Norway PWT TFP Growth vs. Natural Capital Growth Growth

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)
Urban Land	0.787	0.349	0.474	0.312	0.265
	(0.598)	(0.372)	(0.506)	(0.373)	(0.427)
Timber Land	-0.218	-0.306	-0.165	-0.148	-0.449
	(0.186)	(0.189)	(0.090)	(0.129)	(0.277)
Agricultural Land	0.008	0.024	0.002	0.015	0.010
	(0.037)	(0.032)	(0.033)	(0.028)	(0.029)
Forest Cover	0.191	0.192	0.172	0.174	0.103
	(0.147)	(0.161)	(0.069)	(0.080)	(0.168)
Biomass	-0.275	-0.401	0.023	0.052	0.019
	(0.120)	(0.159)	(0.182)	(0.164)	(0.405)
Hydropower	-0.004	-0.000	0.000	0.001	0.001
	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)
Controls for...					
Factor Inputs	No	Yes	No	Yes	Yes
TWFE	No	No	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes
Wald	0.006	0.013	0.264	0.436	0.106
Bootstrap Wald	0.448	0.272	0.909	0.976	–
# Observations	476	476	476	476	476
# Countries	28	28	28	28	28

Note: All independent variables in the table are measured in terms of log changes. The mangroves account is zero for all EU countries and is removed as a covariate for regressions in this table. Controls rows denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. Standard errors clustered at the country level are shown in parentheses. Row "Wald" displays p-values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

USDA ERS Agricultural TFP Finally, we examine whether the renewable capital stocks we consider are correlated with a measure of TFP in the agriculture sector constructed by the USDA's Economic Research Service ([8]). The TFP measure from [8] differs qualitatively from measures in the PWT in that it explicitly accounts for one of the natural capital inputs that we consider, agricultural land, as well as more specialized inputs such as fertilizer use and tillage. As such, we do not expect to see growth in agricultural land as a good predictor when regressing this measure of agricultural TFP growth on the natural capital stocks.

Table 10 shows results when equation 4 in the main text is re-estimated on a panel of agricultural TFP growth for 114 (163) countries between 1997 and 2016 depending on the availability of controls in the PWT. The coefficient on agricultural land area is statistically indistinguishable from zero, as expected given a different metric of agricultural land use is subtracted as an input when the outcome of $\Delta \ln(Ag.\, TFP_{j,t})$ is constructed ([8]). Tests of the joint null hypothesis performed using asymptotic and semiparametric approaches are marginal, and coefficients on individual resources fluctuate substantially depending on the specification we consider.

Table 10: Agricultural TFP Growth vs. Natural Capital Growth

Outcome: $\Delta \ln(Ag.\ TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)
Urban Land	0.301	0.120	0.146	0.078	0.093
	(0.095)	(0.067)	(0.124)	(0.067)	(0.065)
Timber Land	0.164	-0.131	-0.067	-0.284	-0.374
	(0.090)	(0.204)	(0.123)	(0.312)	(0.454)
Agricultural Land	0.027	0.037	0.059	0.039	0.057
	(0.025)	(0.054)	(0.038)	(0.060)	(0.065)
Forest Cover	-0.067	0.157	0.078	0.221	0.276
	(0.094)	(0.189)	(0.121)	(0.255)	(0.308)
Mangroves	0.122	-0.108	0.247	0.426	0.736
	(0.244)	(0.167)	(0.156)	(0.172)	(0.392)
Biomass	0.112	0.072	-0.024	-0.055	0.005
	(0.098)	(0.100)	(0.106)	(0.122)	(0.204)
Hydropower	0.011	-0.012	-0.007	-0.015	-0.016
	(0.004)	(0.008)	(0.008)	(0.009)	(0.009)
Controls for...					
Factor Inputs	No	Yes	No	Yes	Yes
TWFE	No	No	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes
Wald	0.007	0.269	0.376	0.047	0.089
Bootstrap Wald	0.077	0.553	0.355	0.091	–
# Observations	2,808	2,377	3,399	2,377	2,377
# Countries	117	114	163	114	114

Note: All independent variables in the table are measured in terms of log changes. Control rows denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. Row "Wald" displays p-values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Row "Bootstrap Wald" shows p-values from wild bootstrap tests of the joint null. No results are reported for this test in column (5) due to the idiosyncratic trend terms.

3 Producing Törnqvist Quantity Indices for Natural Capital Aggregates

The theory and simulations in the main text consider how the marginal addition of one natural capital to the growth accounting equatoin affects estimates in two stock world. While this in theory corresponds to the case of renewable and non-renewable resources, in reality there multiple forms of natural capital under each of these classifications. Aggregating various non-renewable and renewable resources up to a single figure for each requires the construction of index numbers; summary statistics for the growth in the quantity of all renewable and non-renewable natural capital inputs used in production each year. This section describes the construction of Törnqvist quantity indices for renewable and non-renewable natural capital used to calibrate the simulations in Section 3 of the main text.

To aggregate the sets of renewable and non-renewable resources to single quantities, we form Törnqvist ([16]) quantity indices for each resource. The Törnqvist quantity index for renewable (non-renewable) resources is a harmonic mean over growth in each renewable (non-renewable) inputs that period, with each input weighted by its share of the total value of the full set of renewable (non-renewable) natural capital inputs. The process for constructing Törnqvist indices consists of three steps: (1) constructing quantities used in production and associated rents accruing to each resource in a given year, (2) computing value share weights, and (3) aggregating weighted growth into composite indices. It aligns closely with the procedure used by the World Bank when constructing the official *Changing Wealth of Nations* (CWON) wealth measures ([18]), but performs all calculations excluding resources where data on quantities extracted or rents are proprietary and not made public.

3.1 Natural Capital Use and Rent Data

Data on quantities of resources extracted (for the case of non-renewables) and levels of stocks (for the case of renewables) are drawn directly from the World Bank CWON dataset ([18]). Table 11 shows the 13 non-renewable natural capital stocks that are inputs into the non-renewable Törnqvist quantity index. Non-renewables consist of the three main fossil fuels that comprise roughly four-fifths of global energy use (i.e., coal, oil, and gas), most of the common ferrous and base metals, and a few precious metals used in production.

Table 11: Non-renewable resources used to construct empirical N_1

Resource	Unit
Coal	Metric tons
Gas	Terajoules[9]
Oil	Barrels
Bauxite	Metric tons
Copper	Metric tons
Gold	Metric tons
Iron	Metric tons
Lead	Metric tons
Nickel	Metric tons
Phosphate	Metric tons
Silver	Metric tons
Tin	Metric tons
Zinc	Metric tons

Table 12 shows the seven renewable natural capital stocks that are inputs into the renewable Törnqvist quantity index. These resources align directly with those used as independent variables in the regression analysis listed in Table 2.

Table 12: Renewable resources used to construct empirical N_2

Resource	Unit
Urban land area	Index
Total timber forest	Square Kilometers
Agricultural land area	Square Kilometers
Total forest-cover	Square Kilometers
Mangrove ecosystem area	Hectares
Total biomass	Metric tons
Hydropower electricity generation	Gigawatt-hours

3.2 Data Construction

For each non-renewable resource, quantities and rents (i.e., income attributable to that resource in that year) are taken directly from the CWON dataset at the country-by-year level. Quantities of each resource r extracted in country j in year t are denoted as

$$Q_{j,t}^{(r)}$$

and are based on physical quantities (e.g., barrels of oil produced or tons of minerals mined). This approach considers only the use of non-renewable resources as inputs to extractive industries, where resources in their raw form are transformed into commodities sold as intermediate or final goods. This measure of non-renewable inputs differs substantially from the case where they are intermediate goods in production (e.g., crude oil as an input to oil refining), which are excluded when calculating GDP to avoid double counting. The associated rents are denoted

$$R_{j,t}^{(r)}$$

and are calculated by the World Bank based on on quantities extracted multiplied by local or global commodity market prices ([18]).

For the case of renewables, quantities are constructed in the same manner, but based on physical measurement of stocks, rather than the flow of services provided. As discussed in the main text, this requires assuming that the services renewable natural capital stocks provide are done directly in proportion to physical quantities. Specifically, we require that services provided per unit of stock of a given resource are the same across countries and years. This lets us set the percent changes in renewable service flows each year equal to the percent changes in stocks, an object that is measured in the CWON data. We assign annual rents for renewables based on the flow value imputed to each service in the CWON data. These values are based on various studies that use direct or non-market valuation methods to assess the annual benefits from physical quantities of natural capital stocks. For example, the value of farmland is assigned based on the output of the agriculture it facilitates, while mangroves are valued based on the flood protection they provide ([18]). All resource datasets are merged into a single country-by-year panel covering non-renewable (e.g., fossil fuels, minerals) and renewable (e.g., forests, agricultural land, hydropower) quantities and rents.

3.3 Gross Growth Rates

For each series from the CWON giving the quantity of a given resource used in country j, the gross growth rate is computed as:

$$\hat{Q}_{j,t}^{(r)} = \frac{Q_{j,t}^{(r)}}{Q_{j,t-1}^{(r)}}$$

To avoid leverage from outliers driven by either anomalous growth or administrative errors in data entry, growth values are top-coded at a ratio of 2 (i.e., extraction is assumed to not grow at a rate larger than 100% per annum):

$$\hat{Q}_{j,t}^{(r)} = \min\left(\hat{Q}_{j,t}^{(r)}, 2\right)$$

Between 0 and 265 values for gross growth rates are winsorized for each resource. Observations affected by missing data or zero base-period values are normalized to one. After these adjustments, the dataset has 11,067 entries, a balanced panel of 217 geographic entities between 1970 and 2020.

A majority (between 6,480 and 9,942) of entries are initially missing for all resources and replaced with a gross growth rate of one.

3.4 Törnqvist Weights

Non-renewable resources: Total rents paid to non-renewable natural capital each period in country i can be calculated as

$$R_{j,t}^{NR} = \sum_{r \in \mathcal{N}} R_{j,t}^{(r)}$$

where $\mathcal{N}$ is the set of non-renewable resources given in Table 11. The value share for each resource (its share of total rents across all non-renewable resources that period) is calculated using:

$$w_{j,t}^{(r)} = \frac{R_{j,t}^{(r)}}{R_{j,t}^{NR}}$$

These value shares are used to construct period-t Törnqvist weights based on an equally-weighted average of rent shares in period t and $t-1$

$$\tilde{w}_{j,t}^{(r)} = \frac{1}{2}\left(w_{j,t}^{(r)} + w_{j,t-1}^{(r)}\right)$$

For non-renewable resources, we are able to construct rent weights for 6,064 observations at the country level, an unbalanced panel across all 217 countries.

Renewable resources: Total renewable rent each year is defined analogously as

$$R_{j,t}^{R} = \sum_{k \in \mathcal{R}} R_{j,t}^{(k)}$$

and weights in each year are are computed using:

$$w_{j,t}^{(k)} = \frac{R_{j,t}^{(k)}}{R_{j,t}^{R}}$$

For the construction of this index, urban land is assigned a wealth value equal to that of agricultural land as opposed to one that is 1/1.24 of the produced capital stock following [13]. For renewable resources, we are able to construct rent weights for 3,684 observations at the country level, an unbalanced panel across 149 countries between 1995 and 2020. Because this leaves us with a substantial set of observations where quantities are observed but rent weights are missing, each resource is instead assigned the average value of weights across all years where data are available

$$\bar{w}_{j}^{(k)} = \frac{1}{T} \sum_{t=1}^{T} w_{j,t}^{(k)}$$

for all time when constructing the indices for renewable inputs.

3.5 Törnqvist Indices

Non-renewable resources: For each resource r in the set of non-renewables $\mathcal{N}$, individual gross changes in quantities are then exponentiated:

$$Q_{j,t}^{(r)\,*} = \left(\hat{Q}_{j,t}^{(r)}\right)^{\tilde{w}_{j,t}^{(r)}}$$

The composite Törnqvist quantity index for non-renewable inputs $\mathcal{Q}_{j,t}^{N}$ is the product across resources of all exponentiated terms.

$$\mathcal{Q}_{j,t}^{N} = \prod_{r \in \mathcal{N}} Q_{j,t}^{(r)\,*}$$

Growth in the non-renewable index is then defined as

$$g_{j,t}^{NR} \triangleq \mathcal{Q}_{j,t}^{N} - 1$$

when used for calculating in-sample means, standard deviations, and correlations for non-renewable resource use growth at the national level.

Renewable resources: The renewable Törnqvist index $\mathcal{Q}_{j,t}^{R}$ is constructed in an analogous manner. First, gross quantity changes for all renewable resources k are exponentiated using the time-invariant rent share weights

$$Q_{j,t}^{(k)\,*} = \left(Q_{j,t}^{(k)}\right)^{\bar{w}_{i}^{(k)}}$$

and the index is computed as

$$\mathcal{Q}_{j,t}^{R} = \prod_{k \in \mathcal{R}} Q_{j,t}^{(k)\,*}$$

Growth in the renewable natural capital Törnqvist quantity index is then set as

$$g_{j,t}^{R} \triangleq \mathcal{Q}_{j,t}^{R} - 1$$

when used for calculating correlations at the national level.

Finally, both $g_{j,t}^{R}$ and $g_{j,t}^{NR}$ are topcoded at 1, ruling out over 100% growth in the quantity indices. This affects 86 instances of the renewables index and zero instances of the non-renewables index.

3.6 Regression Analysis

The Törnqvist indices for renewable and non-renewable capital inputs are intended to capture factor input growth the national level. We know from common knowledge that these inputs, especially non-renewable hydrocarbons, should explain some degree of output growth. To ensure growth in our indices for the use of non-renewable and renewable capital are positively associated with output, we run the following regression

$$\Delta \ln(y_{j,t}) = \gamma_1 \; g^{NR}_{j,t} + \gamma_2 \; g^{R}_{j,t} + \sum_{x \in \boldsymbol{Z}} \alpha_z \; g_{z_{j,t}} + \delta_j + \zeta_t + \varepsilon_{j,t} \tag{18}$$

where $y_{j,t}$ is the logarithm of real GDP at constant national prices taken from the PWT intended for use measuring GDP growth ([7]). The set $\boldsymbol{Z} = \{K, L, HC, LS\}$ contains controls for growth in produced capital, employment, human capital, and the labor share in each country each year. Terms (δ_j, ζ_t) are again two-way fixed effects, and $\varepsilon_{j,t}$ are idiosyncratic error terms. The coefficients γ_1 and γ_2—output elasticities for our indices of non-renewable and renewable capital—are our estimands of interest and the feasible analogs of the theoretical counterparts in the simulation section.

Table 13: TFP Growth vs. Natural Capital Growth

$\Delta \ln(y_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
$g^{R}_{j,t}$	0.130	0.073	0.100	0.064	0.066	0.047
	(0.049)	(0.040)	(0.039)	(0.037)	(0.037)	(0.016)
$g^{NR}_{j,t}$	0.025	0.013	0.017	0.011	0.011	0.013
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Controls for…						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
# Observations	4974	3953	4974	3953	3953	3697
# Countries	134	104	134	104	104	104

Note: Standard errors clustered at the country level are shown in parentheses. Rows under the "Controls for …" heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable and is estimated using the Arellano-Bond (1991) dynamic panel estimator.

Table 13 shows the resulting estimates of γ_1, γ_2 from five variants of the specification in equation 18 when estimated on an unbalanced panel of either 104 or 134 countries between 1972 and 2019 (the number of observations and covered countries depends on the availability of controls). Consistent with our priors that growth in the use of natural capital would be associated with output growth, estimates of output elasticities γ are positive across all specifications for growth in both input indices. Point estimates for the output elasticities of non-renewable and renewable natural capital services hover around 0.02 and 0.06 respectively. For non-renewables, the estimates are significant at traditional levels across all specifications ($p < 0.01$) and comparable in magnitude to the output elasticity of energy (which is typically composed mainly of fossil fuels) found in the literature using Cobb-Douglas specifications for aggregate output (see [9, 3]). The coefficients on renewables, while larger in magnitude, are less precise with p-values ranging from 0.01 to 0.13 depending on specification. Both variables also survive a standard 10-fold cross-validation procedure for specification selection. Given the results in Table 13 for $\hat{\gamma}_2$, we assign γ_2 as 0.05 when calibrating the simulations described in Section 4.

4 Country-Level Simulation Exercise

We now describe the simulation exercise underpinning the results in Section 3 of the main text.

4.1 Setup

Data Generating Process for Output Starting with the model of Section 1.4, we again assume output growth in each country is comprised of growth in productivity, labor, produced capital, and two forms of natural capital inputs (non-renewables N_1 and renewables N_2). Joint observations of growth rates for all determinants of output growth are assumed to be independent and identically distributed draws of vector $\boldsymbol{g}_{j,t}$ from a fixed distribution for each country with mean $\boldsymbol{\mu}_j$ and variance Σ_j. Output growth in each simulated country j is generated by

$$g_{Y,j,t} = \boldsymbol{g}'_{j,t}\boldsymbol{\gamma}_j \tag{19}$$

where $\boldsymbol{\gamma}_j \triangleq [1, \gamma_K, \gamma_L, \gamma_1, \gamma_2]'$ is the vector of output elasticities for that country.

Functional Forms and Structural Parameters We assume a log-linear production function with identical output elasticities for natural capital inputs in each country. We set $\gamma_2 = 0.05$ based on the analysis in Section 3.6. In turn, to induce an overall output elasticity of 0.1 for natural capital (following [1]), we set γ_1 as 0.05. This ad-hoc choice is slightly lower than our point estimates in Section 3.6, but in line with estimates of the factor share for fossil energy in the United States since 1947 ([11]).[10]

Procedure for Estimating TFP Growth We assume an econometrician observes a subset of growth in factor inputs—those for K, L, and N_1—along with growth in output without error for 25 periods (years) in each country. We select a sample size of 25 periods because the CWON data on natural capital cover this many years in each country. She deploys the model estimating TFP growth while omitting non-renewable resources (model 1) by estimating:

$$g_Y = \hat{\gamma}_A + \hat{\gamma}_K g_K + \hat{\gamma}_L g_L + \epsilon \tag{20}$$

on her time series of data for growth in $\{K, L\}$ in a given country using ordinary least squares (OLS). Her estimator for average TFP growth under model 1 is the intercept term $\hat{\gamma}_A$. She estimates the analog for the case including non-renewable resources (model 2)

$$g_Y = \tilde{\gamma}_A + \tilde{\gamma}_K g_K + \tilde{\gamma}_1 g_{N_1} + \tilde{\gamma}_L g_L + \nu \tag{21}$$

on time series for growth in $\{K, L, N_1\}$. The OLS-based estimator for average TFP growth under model 2 is $\tilde{\gamma}_A$.

[10] Note we need not make assumptions on γ_L and γ_K in order to solve for the bias or mean squared error of the intercept term.

Data Generating Process for Growth in Inputs We assume at the country level that growth in TFP (g_{TFP}) is uncorrelated with growth in other factors. This restriction can be relaxed to allow arbitrary correlations between growth in factor inputs and true TFP growth, but we elect to set these correlations as zero in all countries as we do not have strong priors for what these unobservable correlations should be. The simulation procedure would remain virtually unchanged in this case apart from assigning alternative values to those off-diagonal terms in Σ_j for each country.

Calculating the bias and mean squared error for the TFP estimators from on models 1 and 2 requires knowing the mean ($\boldsymbol{\mu}_j$) and variance (Σ_j) governing the joint distribution of growth terms $\boldsymbol{g}_j$ in each country.[11] The moments and the sources we use to calibrate them are listed in the first column of Table 14. Columns 2 and 3 of Table 14 list respectively the sample analogs and the source data used to construct them. We set the parameters governing average growth in the use of natural capital equal to the sample means for the growth in renewable and non-renewable Törnqvist indices constructed in Section 3 between 1996 and 2019 (Table 14 rows one and two). Rows three, four, and five—governing expected growth in produced capital services, labor, and TFP—are set equal to their sample analogs in the Penn World Tables ([7]) over the same 1996-2019 period. Finally, higher order moments are set based on the empirical covariances between growth in our Törnqvist non-renewable index and growth in other inputs we take from the PWT during the 1996-2019 period.

Table 14: Simulation Moments and Sources

Population Moment	**Sample Moment**	**Source**
$\mathbb{E}[g_{N_1}]$	$\bar{g}^R$	CWON (Sec. 3)
$\mathbb{E}[g_{N_2}]$	$\bar{g}^{NR}$	CWON (Sec. 3)
$\mathbb{E}[g_K]$	$\bar{g}_K$	PWT
$\mathbb{E}[g_L]$	$\bar{g}_L$	PWT
$\mathbb{E}[g_{TFP}]$	$\bar{g}_{TFP}$	PWT
Σ	$\widehat{\Sigma}$	CWON & PWT

Note: Subscripts indexing each object by country are dropped in the first two columns; all population moments are assigned the sample analogs on a country-by-country basis. Bars denote sample averages while hats denote sample analogs for the VCE matrix Σ. CWON refers to the World Bank *Changing Wealth of Nations* database ([18]), and PWT refers to the Penn World Table ([7]).

Given the limited overlap in data between the CWON and PWT data, we are ultimately able calculate sample analogs of $\boldsymbol{\mu}$ and Σ for 100 countries which we substitute directly into the formulas for bias and RMSE below.

[11] Technically, we need not know $\mathbb{E}[g_{TFP}]$ to calculate the bias or expected MSE. However, we do use this to calculate a ballpark ratio of bias reduction to true underlying growth so it's treated as needed here.

4.2 Bias in Estimated TFP

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We now turn to calculating the bias and mean squared error for each estimator at the country level. As shown in Section 5.4, when g_{TFP} is independent and identically distributed, the bias for the estimate of average TFP growth from model 1 is

$$\mathbb{E}[\hat{\gamma}_A - \mathbb{E}[g_{TFP}]] = \gamma_1 \mathbb{E}[g_{N_1} - \kappa_{N_1,\tilde{K}} g_K - \kappa_{N_1,\tilde{K}} g_L] + \gamma_2 \mathbb{E}[g_{N_2} - \kappa_{N_2,\tilde{K}} g_K - \kappa_{N_2,\tilde{K}} g_L] \tag{22}$$

where the $\kappa_{N_j,\cdot}$ coefficients are generated by the projections of g_{N_1} and g_{N_2} onto the vector of included factor growth $\hat{\boldsymbol{g}} = [g_K, g_L]'$:

$$\begin{aligned} \boldsymbol{\kappa}_{N_1} &= \mathbb{E}[\hat{\boldsymbol{g}}\,\hat{\boldsymbol{g}}']^{-1}\mathbb{E}[\hat{\boldsymbol{g}}\;g_{N_1}] \\ \boldsymbol{\kappa}_{N_2} &= \mathbb{E}[\hat{\boldsymbol{g}}\,\hat{\boldsymbol{g}}']^{-1}\mathbb{E}[\hat{\boldsymbol{g}}\;g_{N_2}] \end{aligned}$$

For the case of Model 2 where only N_2 is omitted, the bias is

$$\mathbb{E}[\tilde{\gamma}_A - \mathbb{E}[g_{TFP}]] = \gamma_2 \mathbb{E}[g_{N_2} - \lambda_{N_2,\tilde{K}} g_K - \lambda_{N_2,\tilde{L}} g_L - \lambda_{N_2,\tilde{N}_1} g_{N_1}] \tag{23}$$

where the $\lambda_{N_2,\cdot}$ coefficients are generated by the projection of g_{N_2} onto $\tilde{\boldsymbol{g}} = [g_K, g_L, g_{N_1}]'$. :

$$\boldsymbol{\lambda} = \mathbb{E}[\tilde{\boldsymbol{g}}\,\tilde{\boldsymbol{g}}']^{-1}\mathbb{E}[\tilde{\boldsymbol{g}}\;g_{N_2}]$$

4.3 Expected Improvements in Root Mean Squared Error

Under the assumption of homoskedastic errors (constant variance in underlying TFP growth across observations) and independent sampling of growth terms, Section 5.5 shows the estimator $\hat{\gamma}$ in (20) has conditional variance

$$\mathrm{Var}(\hat{\gamma}|\hat{\boldsymbol{g}}) = \left(\sigma^2_{TFP} + \begin{bmatrix}\gamma_1, \gamma_2\end{bmatrix} \Sigma_{1,2|\boldsymbol{g}} \begin{bmatrix}\gamma_1 \\ \gamma_2\end{bmatrix}\right) (\hat{\boldsymbol{g}}'\hat{\boldsymbol{g}})^{-1}$$

where here in a slight abuse of notation, $\hat{\boldsymbol{g}} = [1, g_K, g_L]$ is the 25×2 matrix of observed growth terms (and a constant) included under model 1 and $\Sigma_{1,2}$ is the (conditional) covariance matrix for g_{N_1} and g_{N_2}. The conditional variance of of the intercept term is then

$$\mathrm{Var}(\hat{\gamma}_A|\hat{\boldsymbol{g}}) = \frac{\sigma^2_{TFP} + \gamma_1^2\sigma_1^2 + \gamma_2^2\sigma_2^2 + 2\gamma_1\gamma_2\sigma_{1,2}}{25}$$

where $\sigma^2_\cdot$ and $\sigma_{1,2}$ are the (conditional) variances and covariance of growth in the natural capital stocks.

Section 5.5 shows the estimator of TFP from model 2 has variance

$$\mathrm{Var}(\tilde{\gamma}_A|\tilde{\boldsymbol{g}}) = \frac{\sigma^2_{TFP} + \gamma_2^2\sigma_2^2}{25}$$

where $\tilde{\boldsymbol{g}} = [1, g_K, g_L, g_{N_1}]$ is the matrix of regressor under Model 2.

We calculate the expected change in root mean squared error term—Δ RMSE—as

$$\Delta\mathrm{RMSE} \triangleq \sqrt{\mathbb{E}[(\hat{\gamma}_A - \mathbb{E}[g_{TFP}])^2|\hat{\boldsymbol{g}}]} - \sqrt{\mathbb{E}[(\tilde{\gamma}_A - \mathbb{E}[g_{TFP}])^2|\tilde{\boldsymbol{g}}]} \tag{24}$$

which can be decomposed as

$$\Delta\mathrm{RMSE} \triangleq \sqrt{\mathrm{Var}(\hat{\gamma}_A|\hat{\boldsymbol{g}}) + \mathbb{E}[\hat{\gamma}_A - \mathbb{E}[g_{TFP}]|\hat{\boldsymbol{g}}]^2} - \sqrt{\mathrm{Var}(\tilde{\gamma}_A|\tilde{\boldsymbol{g}}) + \mathbb{E}[\tilde{\gamma}_A - \mathbb{E}[g_{TFP}]|\tilde{\boldsymbol{g}}]^2} \tag{25}$$

as $\mathbb{E}[g_{TFP}]$ is a constant and growth in TFP is assumed iid. The (squared) bias terms δ and ε are defined in Section 4.2 and become biases conditional on the observed portion of $\boldsymbol{g}$ when placed inside the operator $\mathbb{E}[\cdot|\boldsymbol{g}]$.

4.4 Constructing the Country-Level Overlay on Figure 1

Table 15 shows data for the set of 100 individual countries overlain on Figure 1 (plotted in Figure 2) in the main text. The two columns labeled bias and RMSE Reduction denote the effect of adopting an estimator including non-renewable resources on the accuracy of the TFP estimate at the country level in unconditional expectation and conditional on observed data.

Table 15: Country-Level Sample Moments and Simulation Results

ISO3 Code	$\bar{g}_{N_1}$	$\bar{g}_{N_2}$	$\hat{\rho}(g_{N_1}, g_K)$	$\hat{\rho}(g_{N_1}, g_{N_2})$	Bias Reduction	RMSE Reduction	$\bar{g}_{TFP}$
AGO	0.04	0.01	0.13	0.09	9.26e-04	1.66e-04	0.02
ARG	-0.01	-0.00	0.13	-0.20	8.36e-04	8.21e-05	-0.00
ARM	0.16	0.01	0.16	0.19	9.02e-03	4.24e-03	0.06
AUS	0.05	-0.01	-0.13	0.08	2.68e-03	1.95e-03	0.00
AUT	-0.02	0.00	-0.08	0.08	-1.77e-04	-2.62e-05	0.00
BDI	0.07	-0.00	0.07	0.08	2.45e-03	1.07e-03	-0.01
BEL	0.20	-0.00	-0.43	0.25	2.60e-02	2.36e-02	0.00
BEN	0.02	0.01	-0.31	-0.08	4.17e-04	3.74e-05	0.02
BFA	0.21	0.00	0.06	-0.10	2.10e-03	6.73e-04	0.01
BGR	-0.00	-0.00	-0.25	0.16	1.30e-03	1.72e-04	-0.01
BHR	0.04	-0.00	0.13	-0.00	2.07e-03	1.11e-03	-0.01
BOL	0.04	-0.00	-0.19	-0.16	1.00e-03	2.95e-04	0.00
BRA	0.06	0.01	0.15	0.10	1.29e-03	4.37e-04	-0.01
BWA	0.01	-0.00	-0.08	-0.21	3.99e-03	1.44e-03	-0.01
CAF	0.08	0.00	0.31	-0.15	3.95e-03	6.71e-04	-0.00
CAN	0.02	0.00	0.13	-0.05	5.54e-04	1.29e-04	0.00
CHE	0.14	0.00	-0.02	-0.49	1.63e-03	1.24e-03	0.01

ISO3 Code	$\bar{g}_{N_1}$	$\bar{g}_{N_2}$	$\hat{\rho}(g_{N_1}, g_K)$	$\hat{\rho}(g_{N_1}, g_{N_2})$	Bias Reduction	RMSE Reduction	$\bar{g}_{TFP}$
CHL	0.03	0.00	0.30	-0.42	-1.49e-03	-5.31e-04	-0.00
CHN	0.04	0.00	0.54	-0.00	7.86e-03	5.51e-03	0.01
CIV	0.09	0.00	-0.12	-0.28	2.04e-03	5.18e-04	0.01
CMR	0.02	0.00	-0.24	0.45	1.69e-02	1.47e-02	0.01
COL	0.02	0.00	0.40	0.04	9.06e-04	2.64e-04	-0.00
CRI	0.18	-0.00	-0.14	-0.05	2.18e-02	1.83e-02	0.01
CZE	-0.01	-0.00	0.23	-0.43	1.28e-03	3.76e-04	0.01
DEU	-0.03	-0.00	0.37	-0.02	3.92e-03	2.69e-03	0.00
DNK	-0.03	-0.00	0.31	-0.54	1.09e-02	9.13e-03	0.00
DOM	0.06	-0.00	0.04	-0.22	1.28e-02	9.78e-03	0.02
ECU	0.01	0.01	0.01	-0.21	-3.37e-05	-5.34e-06	0.00
EGY	-0.00	0.00	0.09	0.38	1.67e-04	3.56e-05	-0.01
ESP	-0.01	-0.00	-0.49	0.13	9.05e-03	8.07e-03	-0.00
EST	0.06	0.00	0.24	0.68	2.90e-03	9.04e-04	0.02
FIN	0.06	-0.00	0.08	0.21	3.93e-03	2.37e-03	0.01
FRA	-0.07	-0.00	0.04	0.22	4.15e-03	3.15e-03	0.00
GAB	-0.03	-0.00	0.10	-0.35	4.61e-04	3.01e-05	-0.01
GBR	-0.05	0.00	0.46	-0.25	7.46e-03	6.49e-03	0.00
GRC	-0.02	-0.01	0.26	-0.11	1.99e-03	8.91e-04	-0.00
GTM	0.04	-0.00	-0.06	0.10	1.13e-02	9.09e-03	0.00
HND	0.02	-0.00	-0.13	0.02	2.07e-03	6.39e-04	-0.00
HRV	-0.01	-0.01	0.07	-0.05	2.85e-03	1.66e-03	0.01
HUN	-0.04	-0.00	-0.06	0.11	1.69e-03	7.02e-04	0.01
IDN	0.01	0.01	0.16	-0.07	-1.26e-04	-1.55e-05	0.00
IND	0.03	-0.00	-0.03	-0.02	9.99e-04	2.22e-04	0.02
IRL	-0.03	0.00	-0.03	0.13	5.24e-04	3.03e-05	0.01
IRN	0.00	-0.01	0.20	-0.07	1.98e-03	3.84e-04	-0.01
IRQ	0.10	-0.00	-0.34	-0.23	3.60e-03	3.83e-04	0.03
ISR	0.10	0.00	-0.59	-0.31	2.60e-02	2.38e-02	0.00
ITA	-0.04	-0.00	-0.07	-0.01	1.52e-03	8.39e-04	-0.01
JAM	0.00	-0.00	0.36	-0.17	2.03e-03	8.30e-04	-0.01
JOR	0.03	-0.00	-0.04	-0.10	1.42e-03	4.61e-04	-0.00
JPN	-0.02	-0.00	-0.10	-0.17	3.42e-04	6.61e-05	0.00
KAZ	0.02	0.00	-0.05	-0.19	2.53e-03	6.85e-04	0.03
KEN	0.16	-0.00	-0.12	-0.03	5.32e-02	4.92e-02	-0.00
KGZ	0.08	-0.00	-0.38	0.23	1.90e-02	1.52e-02	0.02
KOR	-0.08	-0.00	0.07	0.19	7.27e-03	5.23e-03	0.01
KWT	0.03	0.00	0.07	0.24	1.33e-03	1.53e-04	-0.03
LAO	0.19	0.01	-0.04	0.32	3.19e-02	2.91e-02	0.00
LKA	0.01	0.00	0.14	0.20	3.91e-03	2.52e-03	0.00
LTU	0.00	-0.00	-0.19	-0.46	3.32e-03	1.29e-03	0.02
MAR	0.01	-0.00	-0.29	-0.15	1.03e-02	7.34e-03	0.01
MEX	-0.01	-0.00	0.16	0.25	2.03e-03	9.12e-04	-0.00
MNG	0.09	-0.00	-0.06	0.15	3.61e-03	1.53e-03	0.02
MOZ	0.15	0.02	0.02	-0.07	1.50e-02	1.25e-02	0.01
MRT	0.03	-0.00	-0.10	0.37	1.68e-03	2.77e-04	-0.00
MYS	0.01	0.00	0.50	0.07	7.53e-04	1.11e-04	0.01

ISO3 Code	$\bar{g}_{N_1}$	$\bar{g}_{N_2}$	$\hat{\rho}(g_{N_1}, g_K)$	$\hat{\rho}(g_{N_1}, g_{N_2})$	Bias Reduction	RMSE Reduction	$\bar{g}_{TFP}$
NAM	0.02	-0.00	-0.05	0.06	1.48e-03	3.66e-04	-0.00
NER	0.01	0.01	0.06	0.17	2.49e-02	2.09e-02	0.01
NGA	0.01	0.00	0.40	-0.00	1.83e-03	4.02e-04	0.02
NIC	0.11	-0.00	-0.18	-0.07	1.08e-02	8.63e-03	-0.00
NLD	-0.03	-0.00	0.02	0.05	3.29e-03	2.29e-03	0.00
NOR	0.01	0.00	0.03	-0.11	8.26e-03	6.90e-03	-0.00
NZL	0.01	-0.01	0.08	0.07	9.08e-04	4.34e-04	0.00
PAN	0.12	0.01	-0.00	-0.16	1.59e-03	3.13e-04	-0.00
PER	0.05	0.01	-0.39	-0.08	7.50e-03	5.96e-03	-0.00
PHL	0.07	0.00	-0.26	-0.10	9.66e-03	7.87e-03	0.01
POL	0.00	-0.00	0.11	0.18	9.41e-04	3.93e-04	0.02
PRT	-0.00	-0.00	-0.28	-0.04	2.17e-03	1.25e-03	-0.00
QAT	0.06	0.00	0.32	-0.02	-3.65e-04	-1.41e-05	-0.03
ROU	-0.06	-0.00	0.13	-0.25	3.83e-03	1.44e-03	0.02
RUS	-0.03	-0.00	-0.02	-0.07	1.16e-03	1.69e-04	0.02
RWA	0.22	0.01	0.26	-0.54	1.62e-02	1.24e-02	0.03
SAU	0.07	0.00	0.06	-0.30	2.51e-03	1.18e-03	-0.02
SDN	0.14	-0.01	0.25	0.51	4.24e-03	2.01e-03	0.01
SEN	0.09	-0.00	0.26	0.09	3.79e-04	5.59e-05	0.00
SLE	0.09	0.00	0.28	0.01	2.91e-03	3.67e-04	-0.00
SRB	-0.03	-0.00	0.25	-0.28	3.59e-03	1.09e-03	0.03
SVK	-0.03	-0.00	0.11	-0.02	2.49e-03	1.19e-03	0.02
SVN	-0.03	0.01	-0.05	-0.35	9.21e-04	1.65e-04	0.01
SWE	0.03	-0.00	0.11	0.19	-3.11e-05	-2.59e-06	0.01
SWZ	0.05	0.00	0.21	-0.04	3.88e-03	1.83e-03	0.01
TGO	0.09	0.00	-0.25	0.18	1.78e-02	1.32e-02	0.02
THA	0.05	0.00	0.32	-0.24	7.95e-04	8.13e-05	0.02
TTO	0.01	-0.01	-0.05	0.17	5.19e-04	2.10e-05	0.00
TUN	-0.01	0.00	0.34	-0.06	4.74e-03	3.60e-03	0.01
TUR	0.04	-0.00	0.06	0.19	1.60e-03	3.61e-04	-0.00
TZA	0.12	0.00	-0.43	0.07	2.41e-02	2.19e-02	0.01
URY	-0.07	0.01	0.08	-0.05	4.79e-03	3.76e-03	0.01
USA	0.02	-0.00	-0.39	-0.22	3.69e-03	2.99e-03	0.01
ZAF	0.00	-0.00	0.24	0.30	1.33e-03	3.92e-04	-0.00
ZMB	0.04	0.00	0.08	0.16	1.63e-03	4.13e-04	0.02
ZWE	0.00	0.01	0.31	0.39	-6.66e-04	-1.96e-05	0.01

4.5 Cross Sectional Analysis of Simulation Results

This section shows additional cross-sectional analysis of the RMSE improvements from the simulation exercise in Section 4 of the main text. Figure 2 plots RMSE reductions against four additional statistics at the country level measured in 2019: (log) GDP, (log) GDP per capita, human capital levels, and statistical capacity. Panels 2a and 2b indicate that the magnitude of RMSE reductions tends to be decreasing in both measures of income. The relationship is robust and fairly substantial in magnitude; a three-fold increase in GDP decreases estimation error in TFP growth by 20 basis points (double our baseline estimate). A similar relationship holds between these reductions and the level of human capital in a given country (Panel 2c), which is unsurprising given the correlation between education and output. This relationship can be explained by higher-income countries having smaller and less-volatile TFP growth, which lowers the variance of both estimators and thus shrinks the error reduction in absolute terms.

Panel 2d shows the relationship between RMSE improvement and statistical capacity as measured by the World Bank for 67 developing countries in our sample. The correlation here is a fairly precise zero, which we interpret as suggestive evidence that better measurement of natural capital may benefit countries regardless of current national accounting practices. The last two panels take an alternative direction and examines how including non-renewables in our simulations relates to two measures of the *level* of resource intensity in production at the country-level. Panel 2e shows RMSE reductions against resource rents as share of GDP taken from the World Bank's World Development Indicators database ([19]). Panel 2f plots the improvements against an alternative metric for natural capital intensity, resource depletion as a share of gross national income. Perhaps surprisingly, there is a precise negative (albeit small in magnitude) statistical relationship between both measures of resource dependence and error reduction. For highly resource-dependent economies (e.g., Iraq and Kuwait), RMSE reductions are some of the smallest in our sample (1 and 3 basis points respectively). This negative relationship persists qualitatively after controlling for the other factors that have negative correlations shown here (i.e., human capital and GDP), but loses statistical significance and flips signs when conditioned on historical volatility in TFP (the driver of the variance portion of RMSE spoken to above).

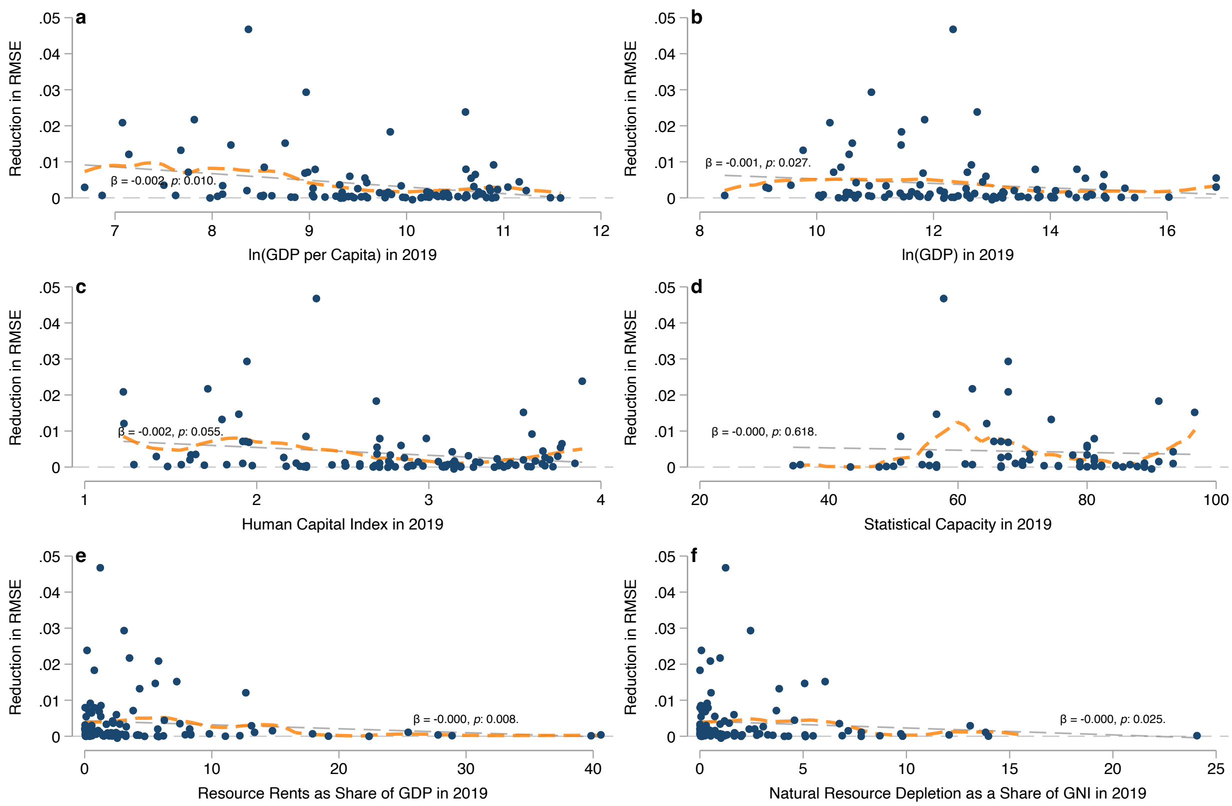

Figure 2: Panels 2a and 2b plot reductions in the RMSE of estimated TFP growth against the (logarithm) of GDP per capita and aggregate GDP at the country level. Panels 2c and 2d plots these improvements against measures of human capital and statistical capacity take from the Penn World Tables ([7]). Panels 2e and 2f show RMSE reductions against the share of GDP that comes from resource rents and the share of national income attributable to natural resource depletion, respectively. Dashed lines show fits from a simple linear regressions and the overlain text shows the $\hat{\beta}$ coefficient from the linear regression along with the associated p-values. Dashed orange lines shows fitted values from a local linear regression ([6]).

5 Proofs

5.1 Proof of Proposition 1

Part 1 Let $\bar{g} \triangleq \max\{\mathbb{E}[g_K], \mathbb{E}[g_L]\}$ and $\underline{g}_N \triangleq \min_j\{\mathbb{E}[g_{N_j}]\}$. Suppose that $\underline{g}_N > \bar{g}$. Then

$$\begin{aligned}\mathbb{E}[\varepsilon] &= [\gamma_K - \hat{\gamma}_K]\mathbb{E}[g_K] + [\gamma_L - (1-\hat{\gamma}_K)]\mathbb{E}[g_L] + \sum_{j=1}^{J}\gamma_j\mathbb{E}[g_{N_j}] \\ &\geq \sum_{j=1}^{J}\gamma_j g_{N_j} - \sum_{j=1}^{J}\gamma_j\bar{g}\end{aligned}$$

where the inequality follows from $\gamma_K + \gamma_L - 1 = \sum \gamma_j$. We then have

$$\mathbb{E}[\varepsilon] \geq [\underline{g}_N - \bar{g}]\sum_{j=1}^{J}\gamma_j \geq 0$$

Finally, note that adding a correctly-measured natural capital term lowers bias, as we reduce the number of positive factor shares γ_j in the summation in the last line which shrinks bias in absolute terms.

Part 2 Let $\underline{g} \triangleq \min\{g_K, g_L\}$ and $\bar{g}_N \triangleq \max_j\{g_{N_j}\}$ and suppose that $\underline{g} > \bar{g}_N$. Then analagous to the presentation above we have

$$\begin{aligned}\mathbb{E}[\varepsilon] &= [\gamma_K - \hat{\gamma}_K]\mathbb{E}[g_K] + [\gamma_L - (1-\hat{\gamma}_K)]\mathbb{E}[g_L] - \sum_{j=1}^{J}\gamma_j\mathbb{E}[g_{N_j}] \\ &\leq \sum_{j=1}^{J}\gamma_j g_{N_j} - \sum_{j=1}^{J}\gamma_j\underline{g} \\ &\leq [\bar{g}_N - \underline{g}]\sum_{j=1}^{J}\gamma_j \leq 0\end{aligned}$$

where in this case adding a correctly-measured NK stock (removing a term from the summation) decreases bias in nominal and absolute terms.

Part 3 Suppose $\hat{\gamma}_K = \gamma_K$. Then the bias is

$$\mathbb{E}[\varepsilon] = \sum_{j=1}^{J}\gamma_j\mathbb{E}[g_L - g_{N_j}] \lesseqgtr 0$$

Alternatively, if $(1-\hat{\gamma}_K) = \gamma_L$ then we have $\hat{\gamma}_K = \gamma_K + \sum\gamma$ which yields

$$\mathbb{E}[\varepsilon] = \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_K - g_{N_j}] \lessgtr 0$$

5.2 Proof of Lemma 1

Proof. We show here the result for a specific case where each input growth term is iid and output elasticities and factor input growth are measured and/or estimated without error. Introducing correlations in growth terms or measurement error adds more potential sources of ambiguity, but is not necessary to show that adding additional natural capital stocks has ambiguous effects on bias. In this case, the biases under models 1 and 2 from equations (Bias 1) and (Bias 2) become

$$\mathbb{E}[\varepsilon|(\hat{\gamma}_K = \gamma_K, \hat{\gamma}_L = \gamma_L)] = \sum_{j=1}^{J} \gamma_j \mathbb{E}[\underbrace{g_{N_j}}_{\lessgtr 0}]$$

$$\mathbb{E}[\delta|(\tilde{\gamma}_K = \gamma_K, \tilde{\gamma}_L = \gamma_L, \tilde{\gamma}_1 = \gamma_1)] = \sum_{j=2}^{J} \gamma_j \mathbb{E}[\underbrace{g_{N_j}}_{\lessgtr 0}]$$

and the magnitude and sign of both biases remains indeterminate.

The mean squared errors of our estimators when all factor shares and growth terms are measured without error are

$$\text{MSE}(1)|(\hat{\gamma}_K = \gamma_K, \hat{\gamma}_L = \gamma_L) = \mathbb{E}\left[\left(\sum_{j=1}^{J} \gamma_j g_{N_j}\right)^2\right]$$

$$\text{MSE}(2)|(\tilde{\gamma}_K = \gamma_K, \tilde{\gamma}_L = \gamma_L, \tilde{\gamma}_1 = \gamma_1) = \mathbb{E}\left[\left(\sum_{j=2}^{J} \gamma_j g_{N_j}\right)^2\right]$$

Evaluating both means squared errors gives

$$\begin{aligned}
\text{MSE}(1)|(\hat{\gamma}_K = \gamma_K, \hat{\gamma}_L = \gamma_L) &= \mathbb{E}\left[\left(\sum_{j=1}^{J} \gamma_j g_{N_j}\right)^2\right] \\
&= \text{Var}\left(\sum_{j=1}^{J} \gamma_j g_{N_j}\right) + \mathbb{E}\left[\varepsilon\right]^2 \\
&= \sum_{i=1}^{J}\sum_{j=1}^{J} \gamma_j \gamma_i \text{Cov}\left(g_{N_i}, g_{N_j}\right) + \mathbb{E}\left[\varepsilon\right]^2
\end{aligned}$$

For the other estimator we have the analog less the included NK stock N_1:

$$\text{MSE}(2)|(\tilde{\gamma}_K = \gamma_K, \tilde{\gamma}_L = \gamma_L, \tilde{\gamma}_1 = \gamma_1) = \sum_{i=2}^{J}\sum_{j=2}^{J} \gamma_j \gamma_i \text{Cov}\left(g_{N_i}, g_{N_j}\right) + \mathbb{E}\left[\delta\right]^2$$

The difference in mean squared error is

$$\text{MSE}(1) - \text{MSE}(2) = \gamma_1^2 \underbrace{\text{Var}(g_{N_1})}_{\geq 0} + \underbrace{2\gamma_1 \sum_{i=2}^{J} \gamma_j \text{Cov}\left(g_{N_1}, g_{N_j}\right)}_{\lesseqgtr 0} + \underbrace{\mathbb{E}[\varepsilon]^2 - \mathbb{E}[\delta]^2}_{\lesseqgtr 0} \lesseqgtr 0$$

Under the assumption of iid growth terms for all natural capital inputs g_{N_j}, the above difference becomes

$$\text{MSE}(1) - \text{MSE}(2) = \gamma_1^2 \underbrace{\text{Var}(g_{N_1})}_{\geq 0} + \underbrace{\mathbb{E}[\varepsilon]^2 - \mathbb{E}[\delta]^2}_{\lesseqgtr 0} \lesseqgtr 0$$

where the inequality is ambiguous unless $\mathbb{E}[\delta] = 0$ in the case where including N_1 makes the growth accounting equation correctly specified.

□

5.3 Proof of Proposition 2

Proof. The result follows directly from equations (12) and (13) as we've assumed that factor growth terms are independent random variables. □

5.4 Biases of OLS-Based TFP Estimators

We present the bias calculations here for the general case where the set of natural capital stocks has cardinality $J \geq 2$. The bias in expected TFP growth when factor shares are estimated using specification in (20) with OLS

$$
\begin{aligned}
\mathbb{E}[\hat{g}_A - \mathbb{E}[g_{TFP}]] &= \mathbb{E}[g_y] - \mathbb{E}[g_y] + [\gamma_K - \hat{\gamma}_K]\mathbb{E}[g_K] + [\gamma_L - \hat{\gamma}_L]\mathbb{E}[g_L] + \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}] \\
&= -\left[\sum_{j=1}^{J} \gamma_j \kappa_{N_j,\tilde{K}} + \kappa_{A,\tilde{K}}\right]\mathbb{E}[g_K] - \left[\sum_{j=1}^{J} \gamma_j \kappa_{N_j,\tilde{L}} + \kappa_{A,\tilde{L}}\right]\mathbb{E}[g_L] + \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j}] \\
&= \sum_{j=1}^{J} \gamma_j \mathbb{E}[g_{N_j} - \kappa_{N_j,\tilde{K}} g_K - \kappa_{N_j,\tilde{L}} g_L] - \kappa_{A,\tilde{K}}\mathbb{E}[g_K] - \kappa_{A,\tilde{L}}\mathbb{E}[g_L]
\end{aligned}
$$

where the κ_{N_j} coefficients are generated by linear projections of g_{N_j} onto the vector $[g_K, g_L]'$. The bias generated by the OLS specification in (21) follows in a similar manner

$$
\begin{aligned}
\mathbb{E}[\tilde{g}_A - \mathbb{E}[g_{TFP}]] = &- \left[\sum_{j=2}^{J} \gamma_j \lambda_{N_j,\tilde{K}} + \lambda_{A,\tilde{K}}\right]\mathbb{E}[g_K] - \left[\sum_{j=2}^{J} \gamma_j \lambda_{N_j,\tilde{L}} + \lambda_{A,\tilde{L}}\right]\mathbb{E}[g_L] \\
&- \left[\sum_{j=2}^{J} \gamma_j \lambda_{N_j,\tilde{N}_1} + \lambda_{A,\tilde{N}_1}\right]\mathbb{E}[g_{N_1}] + \sum_{j=2}^{J} \gamma_j \mathbb{E}[g_{N_j}] \\
&= \sum_{j=2}^{J} \gamma_j \mathbb{E}[g_{N_j} - \lambda_{N_j,\tilde{K}} g_K - \lambda_{N_j,\tilde{L}} g_L - \lambda_{N_j,\tilde{N}_1} g_{N_1}] \\
&- \lambda_{A,\tilde{K}}\mathbb{E}[g_K] - \lambda_{A,\tilde{L}}\mathbb{E}[g_L] - \lambda_{A,\tilde{N}_1}\mathbb{E}[g_{N_1}]
\end{aligned}
$$

where λ_{N_2} are coefficients from a projection of g_{N_2} onto $[g_K, g_L, g_{N_1}]$.

5.5 Variance the of OLS-Based TFP Estimators

Model 1 Split the true data generating process from (19) into vectors of included and excluded regressors

$$
y = Z\gamma_Z + X_o\gamma_o + \varepsilon \tag{26}
$$

where

$$
Z = [\mathbf{1}, g_K, g_L], \qquad X_o = [g_{N_1}, g_{N_2}],
$$

and

$$
\gamma_Z = (\mathbb{E}[g_{TFP}], \gamma_K, \gamma_L)^\top, \qquad \gamma_o = (\gamma_1, \gamma_2)^\top
$$

We assume here that the error term ε—the portion of output growth unexplained by input growth each period—is governed by residual variation in TFP growth such that

$$\mathbb{E}[\varepsilon|\boldsymbol{g}] = 0 \qquad \text{Var}(\varepsilon) = \sigma^2_{TFP}$$

The OLS estimator from regressing y on Z alone is

$$\hat{\gamma} = (Z'Z)^{-1}Z'y. \tag{27}$$

Substituting the true model,

$$\hat{\gamma} = (Z'Z)^{-1}Z'(Z\gamma_Z + X_o\gamma_o + \varepsilon) \tag{28}$$

$$= \gamma_Z + (Z'Z)^{-1}Z'X_o\gamma_o + (Z'Z)^{-1}Z'\varepsilon\gamma_o \tag{29}$$

Define the combined error term as

$$u^{(1)} \triangleq X_o\gamma_o + \varepsilon \tag{30}$$

The variance of $\hat{\gamma}$ conditional on Z is given by the traditional sandwich estimand

$$\text{Var}(\hat{\gamma} \mid Z) = (Z'Z)^{-1}Z'\Omega_1 Z(Z'Z)^{-1} \tag{31}$$

where

$$\Omega_1 \triangleq \text{Var}(u^{(1)} \mid Z)$$

is a covariance matrix of size $n = 25$. Under the assumption of homoskedastic errors and independent sampling we can write the covariance matrix Ω_1 as

$$\Omega_1 = \tau_1^2 I_n$$

where I_n is the identity matrix of size equal to the sample size n and τ_1^2 is a function of the variance of TFP growth σ^2_{TFP} along with the variance of the omitted terms

$$\tau_1^2 = \sigma^2_{TFP} + \text{Var}(\gamma_1 g_{N_1} + \gamma_2 g_{N_2} \mid Z) \tag{32}$$

Letting

$$\Sigma_{12|Z} \triangleq \text{Var}\left(\begin{bmatrix} g_{N_1} \\ g_{N_2} \end{bmatrix} \Bigm| Z\right)$$

we can write the variance term as

$$\text{Var}(\gamma_1 g_{N_1} + \gamma_2 g_{N_2} \mid Z) = \gamma_o^\top \Sigma_{12|Z}\gamma_o \tag{33}$$

Which gives the result in Section 4.3:

$$\text{Var}(\hat{\gamma} \mid Z) = \left(\sigma_{TFP}^2 + \gamma_o^\top \Sigma_{12\cdot Z}\gamma_o\right)(Z'Z)^{-1} \tag{34}$$

Model 2 Rewrite the true data generating process from (DGP) as

$$y = W\gamma_W + g_{N_2}\gamma_2 + \varepsilon \tag{35}$$

where, similar to the procedure for model 1, we assume

$$\mathbb{E}[\varepsilon|\boldsymbol{g}] = 0 \qquad \text{Var}(\varepsilon) = \sigma_{TFP}^2 \qquad W = [\mathbf{1}, g_K, g_L, g_{N_1}], \qquad \gamma_W = (\mathbb{E}[g_{TFP}], \gamma_K, \gamma_L, \gamma_1)^\top.$$

The OLS estimand from regressing y on W is

$$\tilde{\gamma} = (W'W)^{-1}W'y. \tag{36}$$

Substituting gives

$$\begin{aligned} \tilde{\gamma} &= (W'W)^{-1}W'(W\gamma_W + g_{N_2}\gamma_2) && (37)\\ &= \gamma_W + (W'W)^{-1}W'g_{N_2}\gamma_2 + (W'W)^{-1}W'\varepsilon && (38) \end{aligned}$$

Define the error for model 2 as

$$u^{(2)} \triangleq \gamma_2 g_{N_2} + \varepsilon \tag{39}$$

Then the expected variance conditional on W is

$$\text{Var}(\tilde{\gamma} \mid W) = (W'W)^{-1}W'\Omega_2 W(W'W)^{-1} \tag{40}$$

where

$$\Omega_2 \triangleq \text{Var}(u^{(2)} \mid W).$$

Under the assumption of homoskedastic errors and independent sampling we can write the covariance matrix Ω_2 as

$$\Omega_2 = \tau_2^2 I_n,$$

where (analagous to above)

$$\tau_2^2 = \sigma_{TFP}^2 + \gamma_2^2 \text{Var}(g_{N_2} \mid W). \tag{41}$$

Substituting gives

$$\text{Var}(\tilde{\gamma} \mid W) = \left(\sigma^2 + \gamma_2^2 \text{Var}(g_{N_2} \mid W)\right) (W'W)^{-1} \tag{42}$$

as written in Section 4.3.

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